A variational unification of moderate-Knudsen hydrodynamics with the phase-space sheet: the dfmm framework in 1D and 2D, with action-based AMR and variance-gamma stochastic dressing

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Abstract

We develop a variational discretization of moderate-Knudsen hydrodynamics that unifies the dfmm one-dimensional moment scheme with the phase-space sheet picture of collisionless dynamics. The primary variable is the 4×4 phase-space Cholesky factor L over a Lagrangian mass-coordinate mesh in two spatial dimensions; in 1D it reduces to dfmm's existing eight-field scheme. The deterministic action has connection-form structure with the strain-group $GL(d)$ as the structure group: covariant derivatives respect tensor weights, and the Cholesky sector carries a weighted symplectic form $\omega = \alpha^2 \mathrm{d}\alpha \wedge \mathrm{d}\beta$ with α-local Hamiltonian $\mathcal{H}_{\mathrm{Ch}} = -\frac{1}{2}\alpha^2\gamma^2$, generalized to 2D via principal-axis rotation as $\omega_{2D} = \sum_a \alpha_a^2 \mathrm{d}\alpha_a \wedge \mathrm{d}\beta_a + \omega_{\mathrm{rot}}$. The cold-limit reduction is exact: the action collapses to the mass-coordinate dust action of Abel–Hahn–Kaehler, and the Hessian degeneracy at $\gamma \to 0$ is the variational signature of phase-space rank collapse. Stochastic dressing is added via a generalized Kramers–Moyal action whose noise variance is variance-gamma distributed; the shape parameter λ is empirically related to the gamma-distribution shape of compression burst durations, providing a runtime self-consistency check between the deterministic and stochastic sectors. Bayesian remap onto an Eulerian quadtree mesh, computed exactly via the r3d polynomial-moment kernel with curved-element dimension lifting, implements the law of total cumulants and converts coarse-graining entropy production into physical phase-space spreading rather than numerical dissipation. Adaptive mesh refinement is driven by local discrete action error, unifying shock detection, shell-crossing detection, and turbulence-cascade detection in a single criterion. Passive scalar advection is exact at the discrete level by structural identification with parcel labels. The framework extends naturally to two-fluid systems via the dfmm modular cross-coupling operator. We specify the discrete scheme in detail, present a comprehensive validation hierarchy spanning 1D regression against dfmm benchmarks through 2D novelty tests including dust-trapping in vortices and Zel'dovich pancake collapse, and discuss the methods paper structure.

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1 Introduction

Between strictly hydrodynamic and strictly kinetic descriptions of a fluid lies an intermediate regime of considerable astrophysical interest. Post-shock relaxation layers, weakly collisional intracluster plasmas, the cold streams that feed galactic halos, and the multi-stream regions of cosmological dark matter structure all share a feature: a local distribution function that is not Maxwellian but also not arbitrarily structured. Moment closures handle this regime in principle, but their failure modes are typically invisible to the scheme itself. The dfmm framework [1] addresses this by instrumenting a one-dimensional moment scheme so that closure quality is reported as part of the output, with three diagnostics (heat-flux deviation, Cholesky rank-collapse γ, standardized skewness $|s|$) that fire on physically distinct failure modes.

This paper extends dfmm in two directions simultaneously. First, we develop a *variational* formulation in which the deterministic moment evolution arises as the Euler–Lagrange equations of an action with connection-form structure, and the stochastic dressing arises as a generalized Kramers–Moyal MSRJD action with variance-gamma noise vertices. Second, we extend to two spatial dimensions via the natural 4×4 phase-space Cholesky factor over Lagrangian space, with principal-axis decomposition giving a clean structural reduction.

The variational unification has three structural payoffs that are not available in the original dfmm formulation. (i) The cold-limit reduction is exact and analytically clean: the action collapses to the Abel–Hahn–Kaehler mass-coordinate dust action when velocity dispersion vanishes, and the diagnostic $\gamma \to 0$ acquires a sharp variational interpretation as Hessian degeneracy of the deterministic action. (ii) Bayesian remap with the law of total cumulants converts coarse-graining entropy production into physical phase-space spreading rather than numerical dissipation, with exactness through the r3d polynomial-moment kernel. (iii) Passive scalar advection becomes exact at the discrete level by structural identification of tracers with parcel labels, addressing a long-standing pain point of Eulerian schemes in cosmological and ISM applications.

The paper is structured as follows. Section 2 establishes the variable hierarchy: the 4×4 phase-space covariance, its Cholesky factorization, and the strain group as a structure group on the bundle over Lagrangian space. Section 3 develops the deterministic action in connection form, presents the cold-limit reduction, and identifies the Hessian-degeneracy interpretation of phase-space rank collapse. Section 4 develops the generalized Kramers–Moyal stochastic dressing, derives the variance-gamma noise distribution from the gamma-distributed burst-duration empirical structure, and predicts the residual-kurtosis self-consistency check. Section 5 works out the 2D matrix lift via principal-axis decomposition. Section 6 establishes the Bayesian remap with law of total cumulants and the r3d implementation. Section 7 discusses passive scalar advection. Section 8 specifies the discrete scheme in 1D and 2D, including polynomial reconstruction with positivity, discrete Euler-Lagrange equations, action-based AMR, and the implicit Newton iteration. Section 9 extends to two-fluid systems. Section 10 presents the validation hierarchy and test problem specifications. Section 11 summarizes and discusses extensions.

Throughout, we treat the 1D dfmm scheme as the established reference that the 2D extension must reduce to in the appropriate limits, and the test-problem hierarchy reflects this: regression against dfmm benchmarks in 1D mode is a non-negotiable acceptance criterion before 2D claims are made.

2 Framework: the 4×4 phase-space Cholesky over Lagrangian space

2.1 Mass-coordinate Lagrangian space

Lagrangian space $\mathcal{L}$ is a d-dimensional manifold with mass coordinates $\boldsymbol{m}m \in \mathcal{L}$ ($d = 1, 2, 3$). The fundamental spacetime field is the position map

$$\boldsymbol{x}(\boldsymbol{m}, t) : \mathcal{L} \times \mathbb{R} \to \mathbb{R}^d, \tag{1}$$

with deformation gradient $J^i_j = \partial x^i / \partial m^j$ and volume Jacobian $J = \det(J^i_j) = 1/\rho$. Mass per parcel is fixed: $\mathrm{d}m = \rho_0 \mathrm{d}\xi$ in initial conditions, and $\rho(\boldsymbol{x}, t) = \rho_0 / J$ is derived, never stored.

The Lagrangian frame velocity is $\boldsymbol{u} = \dot{\boldsymbol{x}}$ at fixed $\boldsymbol{m}$. The Eulerian velocity gradient is the connection 1-form on Lagrangian space:

$$G^i_j(\boldsymbol{x}, t) = \frac{\partial v^i}{\partial x^j} = \frac{\partial u^i}{\partial m^k} (J^{-1})^k_j. \tag{2}$$

2.2 The 4×4 phase-space covariance

Over each Lagrangian point we attach a phase-space fiber, characterized at second-moment order by the $2d \times 2d$ covariance matrix

$$M = \begin{pmatrix} M_{xx} & M_{xv} \\ M_{xv}^\top & M_{vv} \end{pmatrix}, \qquad M \succeq 0, \tag{3}$$

with each block $d \times d$. Realizability is $M \succeq 0$. The Cholesky factorization $M = LL^\top$ with L block-lower-triangular

$$L = \begin{pmatrix} L_1 & 0 \\ L_2 & L_3 \end{pmatrix}, \quad L_1, L_3 \in \mathbb{R}^{d \times d} \text{ lower-tri (positive diagonal)}, \; L_2 \in \mathbb{R}^{d \times d}, \tag{4}$$

makes realizability manifest: $M_{xx} = L_1 L_1^\top$, $M_{xv} = L_1 L_2^\top$, $M_{vv} = L_2 L_2^\top + L_3 L_3^\top$.

In dfmm's 1D specialization, $L_1 = \alpha$, $L_2 = \beta$, $L_3 = \gamma$, all scalars; the dynamical fields are (α, β) with $\gamma = \sqrt{M_{vv} - \beta^2}$ derived from the EOS. In 2D, L is a 4×4 lower-triangular matrix with 10 independent components.

2.3 The strain group as a structure group

The phase-space covariance transforms as a tensor under the velocity strain group $GL(d)_v$. Each velocity index carries one factor of G:

$$\begin{aligned} \mathcal{D}_t M_{xx} &= \dot{M}_{xx}, \\ \mathcal{D}_t M_{xv} &= \dot{M}_{xv} - G\, M_{xv}, \\ \mathcal{D}_t M_{vv} &= \dot{M}_{vv} - G\, M_{vv} - M_{vv} G^\top, \end{aligned} \tag{5}$$

where $\dot{} = \partial_t$ at fixed $\boldsymbol{m}$. In tensor language, M_{xx} has weight $(0,0)$ under $GL(d)_v$, M_{xv} has weight $(0,1)$, M_{vv} has weight $(1,1)$. Liouville evolution under collisionless free streaming is

$$\mathcal{D}_t M_{xx} = M_{xv} + M_{xv}^\top, \qquad \mathcal{D}_t M_{xv} = M_{vv}, \qquad \mathcal{D}_t M_{vv} = 0. \tag{6}$$

Right-hand sides are algebraic; all strain coupling is absorbed into the covariant derivative.

2.4 Specialization to 1D

With $d = 1$, $G = \partial_x u$, weights become integer charges $q \in \{0, 1, 2\}$. Charges of the Cholesky factors: $\alpha = L_1$ has charge 0, $\beta = L_2$ has charge 1, $\gamma = L_3$ has charge 1. The covariant derivative is

$$\mathcal{D}_t^{(q)} \phi = \dot{\phi} + q(\partial_x u)\phi. \tag{7}$$

The dfmm Cholesky equations [1] become

$$\mathcal{D}_t^{(0)} \alpha = \beta, \qquad \mathcal{D}_t^{(1)} \beta = \frac{\gamma^2}{\alpha} = \frac{M_{vv}(J, s) - \beta^2}{\alpha}, \tag{8}$$

recovering the bare-derivative form

$$\dot{\alpha} = \beta, \qquad \dot{\beta} + (\partial_x u)\beta = \gamma^2/\alpha. \tag{9}$$

2.5 Liouville invariant

Under collisionless dynamics, the per-parcel phase-space volume $\sqrt{\det M} = \det L$ is conserved. In 1D this is $\alpha\gamma = \det L$. In 2D it's the determinant of the 4×4 Cholesky factor. Mass per parcel is fixed by definition, so the full phase-space volume invariant per unit reference mass is exactly $\det L$. This is the Liouville invariant we use to characterize the deterministic dynamics and to bound the Bayesian remap's entropy-production behavior (Section 6).

3 The deterministic action in connection form

3.1 Cholesky sector: weighted symplectic structure

The dfmm Cholesky flow on (α, β) is *not* Hamiltonian under the canonical symplectic form $\mathrm{d}\alpha \wedge \mathrm{d}\beta$; one finds the integrability condition fails. It *is* Hamiltonian under the weighted form

$$\omega_{1D} = \alpha^2\, \mathrm{d}\alpha \wedge \mathrm{d}\beta, \tag{10}$$

with α-local Hamiltonian density

$$\mathcal{H}^{1D}_{\mathrm{Ch}}(\alpha,\beta;J,s) = -\tfrac{1}{2}\,\alpha^2\,\gamma^2 = -\tfrac{1}{2}\,\alpha^2\big(M_{vv}(J,s)-\beta^2\big). \tag{11}$$

The factor $\alpha^2 = M_{xx}$ in ω_{1D} is precisely the gauge weight of the symplectic potential under the strain group: ω_{1D} has total charge $0+1=1$, which is the charge needed for Hamilton's equations to produce charge-correct covariant time derivatives.

Verification. Hamilton's equations for the weighted form $\omega = f\,\mathrm{d}\alpha\wedge\mathrm{d}\beta$ with Hamiltonian $\mathcal{H}$ are

$$\mathcal{D}_t\alpha = \frac{1}{f}\frac{\partial\mathcal{H}}{\partial\beta}, \qquad \mathcal{D}_t\beta = -\frac{1}{f}\frac{\partial\mathcal{H}}{\partial\alpha}. \tag{12}$$

With $f=\alpha^2$ and $\mathcal{H}_{\mathrm{Ch}}$ from (11): $\partial\mathcal{H}_{\mathrm{Ch}}/\partial\beta = \alpha^2\beta$ gives $\mathcal{D}_t\alpha=\beta$; $\partial\mathcal{H}_{\mathrm{Ch}}/\partial\alpha = -\alpha\gamma^2$ gives $\mathcal{D}_t\beta = \gamma^2/\alpha$. Both equations match (8). ✓

Symplectic potential. A 1-form θ with $\mathrm{d}\theta = \omega_{1D}$ is e.g. $\theta = (\alpha^3/3)\,\mathrm{d}\beta$, giving the Hamilton–Pontryagin action density

$$\mathcal{L}_{\mathrm{Ch}} = \frac{\alpha^3}{3}\,\mathcal{D}^{(1)}_t\beta - \mathcal{H}_{\mathrm{Ch}}. \tag{13}$$

The strain coupling $-(\partial_x u)\beta$ in (9) is generated entirely by $\mathcal{D}^{(1)}_t\beta$ in (13); $\mathcal{H}_{\mathrm{Ch}}$ contains no strain term. The discrete scheme will treat the strain as a parallel-transport operator, not a non-local source.

Note on the symplectic-potential sign convention. The discrete Euler–Lagrange machinery in §8 is sign-sensitive: applying the standard EL convention $\delta\mathcal{L}/\delta q - \mathrm{d}/\mathrm{d}t\,(\delta\mathcal{L}/\delta\dot q) = 0$ to (13) reproduces the boxed Hamilton equations only if the symplectic potential is taken as $\theta = -(\alpha^3/3)\,\mathrm{d}\beta$ (equivalently, the wedge order in ω_{1D} is reversed). The Milestone-1 implementation in `src/cholesky_sector.jl` (header lines 36–43) writes $\mathcal{L}_{\mathrm{Ch}} = -(\alpha^3/3)\,\mathcal{D}^{(1)}_t\beta - \mathcal{H}_{\mathrm{Ch}}$ for this reason. The two conventions describe the same physics; downstream EL machinery must use the matched convention. This discrepancy with the v2 design note (Track A `design/03_action_note_v2.tex` §3.1) is recorded here for archival reference.

3.2 Bulk and entropy sectors

Bulk position $\boldsymbol{x}$ and specific entropy s are scalars (charge 0) under the strain group. The full deterministic action density is

$$\mathcal{L}_{\mathrm{det}} = \tfrac{1}{2}|\dot{\boldsymbol{x}}|^2 + \mathcal{L}_{\mathrm{Ch}} + \mathcal{L}_{\mathrm{dev}} + \lambda_Q\big(\dot Q + Q/\tau - \mathcal{S}_Q\big), \tag{14}$$

with $\mathcal{L}_{\mathrm{dev}}$ the deviatoric stress sector (§3.3) and the heat-flux constraint enforced as a hard Lagrange multiplier (soft-constraint form deferred to §**??**).

Variation of $\boldsymbol{x}$ gives the bulk momentum equation

$$\ddot{\boldsymbol{x}} = -\partial_{\boldsymbol{m}}P_{xx} \quad\Longleftrightarrow\quad \rho\,\mathrm{D}\boldsymbol{u}/\mathrm{D}t = -\partial_{\boldsymbol{x}}P_{xx}, \tag{15}$$

the standard Euler equation. Variation of s gives parcel-wise entropy conservation $\dot s = 0$ in the deterministic adiabatic limit.

3.3 Deviatoric stress as a dynamical variable

In 1D the only relevant deviatoric DOF is the $P_{xx}/P_\perp$ split. In 2D and 3D the deviatoric stress is a traceless symmetric tensor

$$\pi_{ij} = M_{vv,ij} - \frac{1}{d} M_{vv,kk}\delta_{ij}, \tag{16}$$

with Lagrangian-frame evolution

$$\mathcal{D}_t \pi = -\pi/\tau_\pi - 2\eta\, S^{\mathrm{dev}} + \mathcal{Q}(\pi, S), \tag{17}$$

where S^{dev} is the deviatoric strain rate, η a viscosity coefficient, τ_π a relaxation timescale, and $\mathcal{Q}$ collects nonlinear couplings (Burnett-level terms). This is the Israel–Stewart structure with its own Cholesky factorization for realizability. In 1D, π degenerates to a single scalar $\Pi = P_{xx} - P_\perp$ relaxing under BGK at rate $1/\tau$.

3.4 Cold-limit reduction

At $s \to -\infty$: $M_{vv} \to 0$, $\beta \to 0$, $\gamma \to 0$, $\Pi \to 0$. $\mathcal{H}_{\mathrm{Ch}} \to 0$, $\mathcal{L}_{\mathrm{dev}} \to 0$, and the action collapses to

$$S_{\mathrm{cold}} = \int \mathrm{d}t \int \mathrm{d}^d m\, \tfrac{1}{2}|\dot{\boldsymbol{x}}|^2, \tag{18}$$

the mass-coordinate pressureless-dust action of Abel–Hahn–Kaehler. EL equations: $\ddot{\boldsymbol{x}} = 0$, ballistic free streaming. Adding self-gravity is a single term $-\int \mathrm{d}m\, \Phi(\boldsymbol{x}(\boldsymbol{m}, t), t)$ with Poisson's equation as a constraint; the phase-space sheet tessellation framework follows directly. ✓

3.5 Hessian degeneracy at $\gamma \to 0$

The symplectic form $\omega_{1D} = \alpha^2\, \mathrm{d}\alpha \wedge \mathrm{d}\beta$ remains non-degenerate as long as $\alpha > 0$. But the Hessian of $\mathcal{H}_{\mathrm{Ch}}$ on the (α, β) tangent space is

$$\mathrm{Hess}(\mathcal{H}_{\mathrm{Ch}}) = \begin{pmatrix} -\gamma^2 & 2\alpha\beta \\ 2\alpha\beta & \alpha^2 \end{pmatrix}, \quad \det = -\alpha^2(\gamma^2 + 4\beta^2) \xrightarrow{\gamma\to 0, \beta\to 0} 0. \tag{19}$$

At $\gamma = 0$ *and* $\beta = 0$ (rank-one phase-space ridge with no shear), the Hessian is rank-1. This is the variational signature of phase-space rank collapse: the deterministic action loses linear-stability structure at the caustic. The stochastic dressing of §4 re-opens the gap by adding a positive-definite noise vertex.

Note on the magnitude direction at the caustic. At the caustic in the *evolved* state, M_{vv} grows for adiabatic $\Gamma > 1$ (because $J \to 0$ and $M_{vv} = J^{1-\Gamma} e^{s/c_v}$), so γ^2 and $|\det \mathrm{Hess}|$ have spatial *maxima* at the caustic location, not minima. The cold-limit *initial* state has $\gamma \to 0$ uniformly; the *evolved* caustic is where $\beta \to \sqrt{M_{vv}}$ saturates on the trajectory's level set while M_{vv} itself diverges. The variational signature of the caustic is therefore the spatial *shape* — a symmetric peak at the rank-1 ridge — not the magnitude direction. This was confirmed empirically in M1 Phase 3 and Phase 6 (see `reference/figs/phase3_hessian_degen.png` and `reference/figs/A2_cold_sinusoid_density.png`).

3.6 Heat flux gauge structure

The heat flux $Q_{ijk} = \langle (v-u)_i (v-u)_j (v-u)_k \rangle$ has three velocity indices, hence weight $(1,1,1)$ under the strain group. The covariant derivative is

$$\mathcal{D}_t Q_{ijk} = \dot{Q}_{ijk} - G^a_i Q_{ajk} - G^a_j Q_{iak} - G^a_k Q_{ija}. \tag{20}$$

In 1D this reduces to $\mathcal{D}_t^{(3)} Q = \dot{Q} + 3(\partial_x u) Q$, matching dfmm's existing scalar treatment under appropriate relabeling. In 2D, Q_{ijk} has 4 independent components after symmetrization and the gauge structure is non-trivial.

4 Stochastic dressing: variance-gamma from intermittent compression

4.1 Empirical foundation

Wave-pool calibration [4] establishes three empirical facts that constrain the stochastic dressing:

1. Compression burst durations are gamma-distributed with shape parameter $k \approx 2.5$–2.8, robust across compression thresholds (Kolmogorov-Smirnov rejects exponential at $p \to 0$).
2. Conditioned on compressive flow, the closure-error residual $\eta = (\epsilon - b\,\mathrm{d}t)/\sqrt{\sigma^2 \mathrm{d}t}$ is well-fit by a variance-gamma distribution with shape $\lambda \approx 1$–1.6 (production calibration vs. small-data fit).
3. The drift coefficient $C_A \approx 0.34$ (production) decomposes as $C_A = b_{\text{Itô}} + b_{\text{closure}}$; the split requires the Lagrangian-frame density disambiguation that small-data calibration lacks.

4.2 Variance-gamma from variance mixing

Drop the assumption that burst durations are exponential. Compression bursts arrive as a Poisson process at rate λ_b, each with random duration $T_b \sim \Gamma(k, \theta_T)$ and intensity κ_b. Noise variance accumulated in a save interval is

$$V = C_B^2 \sum_{n=1}^{N(T_{\text{save}})} \kappa_n T_n, \quad N(T_{\text{save}}) \sim \text{Poisson}(\lambda_b T_{\text{save}}), \tag{21}$$

which is gamma-distributed at fixed κ, modified to variance-gamma by the multiplicative cascade structure of κ.

Theorem (variance-gamma from variance mixing). If $\epsilon | V \sim \mathcal{N}(0, V)$ with $V \sim \Gamma(\lambda, \theta)$ in the Distributions-canonical (variance-mixture) parametrization ($\mathbb{E}[V] = \lambda\theta$, $\mathrm{Var}(\epsilon) = \lambda\theta$), then $\epsilon \sim \mathrm{VG}(\lambda, \theta)$ with density

$$f(\epsilon) = \sqrt{\frac{2}{\pi}}\, \theta^{-\lambda} \frac{\left(|\epsilon|\sqrt{\theta/2}\right)^{\lambda - 1/2} K_{\lambda - 1/2}\left(|\epsilon|\sqrt{2/\theta}\right)}{\Gamma(\lambda)}, \tag{22}$$

with K_ν the modified Bessel function of the second kind. Direct quadrature confirms $\int f(\epsilon)\,\mathrm{d}\epsilon = 1$ (verified in M1 Track D, `src/stochastic.jl::pdf_variance_gamma` and `test_vg_sampler.jl`). The textbook generalised-hyperbolic form is recovered under the rescaling $\theta_{\text{GH}} = 2\theta$. Special cases: $\lambda = 1$ gives Laplace; $\lambda \to \infty$ (with θ scaled) gives Gaussian. Excess kurtosis is $3/\lambda$.

4.3 The MSRJD action with variance-gamma vertices

The stochastic-action contribution generalizes from quadratic to the variance-gamma cumulant generating function:

$$\mathcal{K}_{\mathrm{VG}}(\hat{\epsilon}; V_{\mathrm{typ}}) = -\lambda \log\bigl(1 - \theta\hat{\epsilon}^2/2\bigr) = \tfrac{1}{2}\lambda\theta\hat{\epsilon}^2 + \frac{1}{4}\lambda\theta^2\hat{\epsilon}^4 + O(\hat{\epsilon}^6). \tag{23}$$

The leading term is the Gaussian noise vertex (recovering MSRJD at quadratic order); higher-order vertices encode the heavy tails. For empirical $\lambda \sim 1$–2, only the quartic correction is numerically relevant.

4.4 Self-consistency check

The framework predicts a measurable consistency relation between the deterministic and stochastic sectors:

$$\boxed{\lambda_{\mathrm{residual}} = (\text{gamma-shape of compression burst durations})} \tag{24}$$

This is a runtime self-consistency check: the implementation monitors both burst-duration shape (from compression-detector statistics) and residual kurtosis (from closure-error vs. deterministic prediction), and warns if they disagree. This is a closure-quality diagnostic on the *stochastic* dressing, parallel to the dfmm closure diagnostics on the deterministic moments.

4.5 2D extension: noise per compressive principal axis

In 2D the strain rate tensor $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ has principal-axis decomposition with eigenvalues $\lambda_1 \geq \lambda_2$ and eigenvectors $\hat{e}_1, \hat{e}_2$. The prediction is independent variance-gamma noise on each compressive principal axis:

$$\sigma_a^2 = C_B^2 \max(-\lambda_a, 0), \quad a = 1, 2. \tag{25}$$

Pure incompressible 2D shear flow has $\lambda_1 = -\lambda_2 \neq 0$, giving *one* compressive direction only — the framework predicts noise on one axis, not both. This is a falsifiable signature for any 2D wave-pool calibration; it distinguishes the principal-axis noise prediction from any trace-based or isotropic alternative.

4.6 Drift decomposition

The drift coefficient $C_A \approx 0.34$ from the wave-pool calibration decomposes as $C_A = b_{\text{Itô}} + b_{\text{closure}}$, with $b_{\text{Itô}} = \frac{1}{2}\partial_u\sigma^2/\sigma^2 \cdot |\sigma|$ the Itô-Stratonovich shift and b_{closure} the genuine mean closure error. The split (predicted $\sim 0.15 + \sim 0.19$) is testable from production calibration data with the Lagrangian-frame density disambiguation. The framework accepts whatever decomposition emerges; the structural claim that C_A has these two components is robust.

5 The 2D matrix lift via principal-axis decomposition

5.1 Why direct tensor pairing fails

The natural conjecture from the 1D form is

$$\omega_{2D}^{\mathrm{naive}} \stackrel{?}{=} (\det L_1)^2 \ \mathrm{Tr}(\mathrm{d}L_1 \wedge \mathrm{d}L_2^{\top}), \tag{26}$$

with L_1 the lower-tri Cholesky of M_{xx}. Working out the 2D Liouville equations for the entries of L_1, L_2 reveals that the off-diagonal entry of L_2 couples to L_1 entries through ratios that the naive trace pairing does not reproduce. The off-diagonal entry is dynamical (it generates cross-axis coupling) but the trace pairing leaves it algebraic. The naive form is incomplete.

5.2 Principal-axis rotation as the structural fix

Work in the local principal-axis basis of M_{xx}: rotate to coordinates where L_1 is diagonal, $L_1 = \mathrm{diag}(\alpha_1, \alpha_2)$. The two principal axes carry independent 1D Cholesky structures, and cross-axis coupling enters via the Maurer–Cartan 1-form of the rotation $R(t) \in SO(2)$:

$$\omega_{2D} = \alpha_1^2\, \mathrm{d}\alpha_1 \wedge \mathrm{d}\beta_1 + \alpha_2^2\, \mathrm{d}\alpha_2 \wedge \mathrm{d}\beta_2 + \omega_{\mathrm{rot}}(\alpha_a, \beta_a, \theta_R), \tag{27}$$

where θ_R is the rotation angle of the principal-axis frame and ω_{rot} is the Berry-curvature-like contribution.

The first two terms are exactly two copies of the 1D weighted symplectic form; reduction to 1D (set $\alpha_2 = \alpha_1$, $\beta_2 = \beta_1$, $\theta_R = 0$) recovers ω_{1D} correctly.

The Berry-connection 1-form Θ_{rot} generating $\omega_{\mathrm{rot}} = d\Theta_{\mathrm{rot}}$ is, in the principal-axis-diagonal restriction,

$$\Theta_{\mathrm{rot}} = \tfrac{1}{3}\big(\alpha_1^3\, \beta_2 - \alpha_2^3\, \beta_1\big)\, d\theta_R, \tag{28}$$

giving the explicit 2-form

$$\omega_{\mathrm{rot}} = \big(\alpha_1^2 \beta_2\, d\alpha_1 + \tfrac{\alpha_1^3}{3}\, d\beta_2 - \alpha_2^2 \beta_1\, d\alpha_2 - \tfrac{\alpha_2^3}{3}\, d\beta_1\big) \wedge d\theta_R. \tag{29}$$

The form is bilinear-antisymmetric under axis swap ($1 \leftrightarrow 2$), vanishes on the iso-diagonal subspace $\alpha_1 = \alpha_2$, $\beta_1 = \beta_2$ (the principal-axis-degenerate gauge boundary), and dimensionally matches the 1D Hamilton-Pontryagin factor $\alpha^3/3$ per axis. Symbolic verification ($d\omega_{2D} = 0$, per-axis Hamilton equations, iso-pullback Berry block, kernel-direction analysis) is in `scripts/verify_berry_connection.py` of the implementation repository.

Normalization. On the iso-diagonal subspace, the per-axis sum $\alpha_1^2\, d\alpha_1 \wedge d\beta_1 + \alpha_2^2\, d\alpha_2 \wedge d\beta_2$ pulls back to $2\alpha^2\, d\alpha \wedge d\beta$ (each diagonal half contributes equally). To recover $\omega_{1D} = \alpha^2\, d\alpha \wedge d\beta$ exactly under the iso reduction, one of two normalization conventions is consistent: (a) carry an explicit $1/2$ in the per-axis weight, $\omega_{2D} = \frac{1}{2}\sum_a \alpha_a^2\, d\alpha_a \wedge d\beta_a + \omega_{\mathrm{rot}}$, or (b) interpret the iso pullback as a projection counting each axis half-weight on the diagonal subspace. Both are internally consistent; this paper adopts convention (a) for explicitness in subsequent work.

5.3 Why this is the right structural choice

- **1D reduction is automatic.** The 1D scheme is a single principal-axis specialization of the 2D action; no separate 1D code path needed.
- **Noise structure aligns naturally.** The variance-gamma noise prediction (per compressive principal axis) lives on the same principal-axis decomposition as the symplectic form. Deterministic and stochastic sectors are both axis-diagonal in this frame.
- **Rotation angle is a physical observable.** It tracks how the local M_{xx} ellipsoid orients itself relative to the strain eigenframe. At shell crossing, principal axes align with the caustic direction and ω_{rot} carries the rotation that brings L_1 to a singular limit.
- **3D extension is straightforward.** Replace $SO(2)$ with $SO(3)$ and three principal axes; same structure with three diagonal 1D-form contributions plus a Berry curvature on $SO(3)$.

5.4 Hamiltonian and equations of motion

The 2D Cholesky Hamiltonian on the principal-axis basis is

$$\mathcal{H}_{\mathrm{Ch}}^{2D} = -\tfrac{1}{2}\big(\alpha_1^2\gamma_1^2 + \alpha_2^2\gamma_2^2\big) + \mathcal{H}_{\mathrm{rot}}(\theta_R), \tag{30}$$

with the Berry-connection contribution to the rotation Hamiltonian $\mathcal{H}_{\mathrm{rot}}$ specified by the Poisson-manifold integrability constraint below. Hamilton's equations on the diagonal sectors give two independent copies of the 1D dfmm Cholesky evolution, one per principal axis. The rotation sector evolves on $SO(2)$ driven by the misalignment between the M_{xx} principal axes and the strain principal axes.

Poisson-manifold constraint on $\mathcal{H}_{\mathrm{rot}}$. In the principal-axis-diagonal restriction, ω_{2D} is a rank-4 Poisson form on the 5D phase space $(\alpha_a, \beta_a, \theta_R)$, with a 1D gauge-direction kernel. Hamilton's equations $\Omega\, X = -dH$ have solutions iff dH is orthogonal to the kernel. This integrability condition fixes $\mathcal{H}_{\mathrm{rot}}$ in the no-strain-misalignment sector (where $\tilde{S}_{12} = 0$) algebraically:

$$\left.\frac{\partial \mathcal{H}_{\mathrm{rot}}}{\partial \theta_R}\right|_{\mathrm{constraint}} = \frac{\gamma_1^2\,\alpha_2^3}{3\alpha_1} - \frac{\gamma_2^2\,\alpha_1^3}{3\alpha_2} - (\alpha_1^2 - \alpha_2^2)\,\beta_1\beta_2. \tag{31}$$

The strain-driven part of $\mathcal{H}_{\mathrm{rot}}$ (which carries the methods-paper §5.5 "misalignment with strain principal axes" coupling) requires lifting the diagonal restriction; this is treated when the off-diagonal L_2 sector is added (see the implementation notes for the 2D-extension paper).

6 Bayesian remap with the law of total cumulants

6.1 Setup

The Lagrangian mesh deforms with the flow; periodic projection to a fixed Eulerian quadtree mesh is needed for diagnostics, force evaluation (gravity), and coarsening. Standard remaps (mass-weighted linear interpolation) lose information about sub-cell velocity spread; the Bayesian remap captures this via the law of total cumulants and converts what would be numerical dissipation into physical phase-space spreading.

6.2 Geometric overlap and polynomial moment transfer

For each Lagrangian cell T_i (deformed) and Eulerian cell C_j, the polygonal overlap $T_i \cap C_j$ is computed exactly, and the polynomial moments of the field reconstructions $\boldsymbol{x}(\boldsymbol{q}), \boldsymbol{u}(\boldsymbol{q}), L(\boldsymbol{q}), s(\boldsymbol{q}), Q(\boldsymbol{q})$ are integrated over the overlap region. For curved-element reconstructions (cubic or higher), we use the dimension-lifting approach: each curved-edge constraint is lifted to a coordinate hyperplane in extended space (typically 5D for triangle–quad overlap with cubic edges), where the overlap region is a polytope and the r3d [2] polynomial-moment kernel gives exact moments.

Implementation via HierarchicalGrids. In the dfmm-2d implementation, the geometric backbone of the remap is provided by the dimension-generic `HierarchicalGrids.jl` [5] (HG) substrate, which hosts the r3d polynomial-moment kernel through its Julia port `r3djl` [6]. The Lagrangian and Eulerian meshes appear to the remap driver as a `PairedMesh` (or more generally a `CompositeMesh`) of an HG `SimplicialMesh` and an HG `HierarchicalMesh`; per-cell field reconstructions are stored as a `PolynomialFieldSet` in the Bernstein basis at order p. The two primitives that do the

geometric work are `compute_overlap` (returns the $\{T_i \cap C_j, m_{ij}, \text{moment integrals}\}$ table) and `polynomial_remap` (assembles the new Eulerian-cell field from those overlap moments via the law of total cumulants specified in §6.3). The overlap cache is invalidated automatically when either mesh is refined or coarsened. The mathematical content of the remap — priors, likelihoods, and posterior moment matching specified in the remainder of this section — is unchanged by this implementation choice; HG and r3djl supply the conservative geometric transfer, and dfmm specifies the Bayesian moment-matching rule on top.

6.3 Law of total covariance

For new Eulerian cell C_j receiving mass fractions w_{ij} from source Lagrangian cells T_i:

$$m_j^{\text{new}} = \sum_i m_{ij}, \tag{32}$$

$$\boldsymbol{u}_j^{\text{new}} = \sum_i w_{ij} \boldsymbol{u}_i, \tag{33}$$

$$M_j^{\text{new}} = \underbrace{\sum_i w_{ij} M_i}_{\text{averaged}} + \underbrace{\sum_i w_{ij} \left(\Delta\phi_i\right)\left(\Delta\phi_i\right)^\top}_{\text{coarse-graining contribution}}, \tag{34}$$

where $\phi = (\boldsymbol{x}, \boldsymbol{u})$ is the phase-space mean and $\Delta\phi_i = \phi_i - \phi_j^{\text{new}}$. The block structure of the coarse-graining contribution is:

- $\Delta\boldsymbol{x}\,\Delta\boldsymbol{x}^\top$ contribution to M_{xx}^{new}: geometric, automatically absorbed by the larger extent of the new Eulerian cell.
- $\Delta\boldsymbol{x}\,\Delta\boldsymbol{u}^\top$ contribution to M_{xv}^{new}: informative, captures the cross-correlation between source-cell mean positions and velocities.
- $\Delta\boldsymbol{u}\,\Delta\boldsymbol{u}^\top$ contribution to M_{vv}^{new}: informative, captures velocity-spread variance from coarse-graining.

The two informative contributions are precisely what standard remaps lose to numerical dissipation; the Bayesian remap converts them into physical phase-space spreading.

6.4 Higher cumulants

The law of total cumulants extends to higher moments. For the heat flux Q at third moment:

$$\begin{aligned} Q_j^{\text{new}} = \sum_i w_{ij} Q_i + \sum_i w_{ij} (\Delta\boldsymbol{u}_i) \otimes M_{vv,i} + \text{symms.} \\ + \sum_i w_{ij} (\Delta\boldsymbol{u}_i)^{\otimes 3}, \end{aligned} \tag{35}$$

appropriately symmetrized. Higher cumulants follow the same pattern.

6.5 Liouville behavior

Theorem (Liouville monotonicity). Under the Bayesian remap, the per-parcel phase-space volume $\det L = \sqrt{\det M}$ is monotonically non-decreasing, with equality if and only if the remap is trivial (one-to-one).

Sketch. The Minkowski determinant inequality gives $\det(\sum w_i M_i)^{1/(2d)} \geq \sum w_i (\det M_i)^{1/(2d)}$, with equality iff all M_i equal. The coarse-graining contribution $\sum w_i (\Delta\phi_i)(\Delta\phi_i)^\top$ is positive semidefinite, adding to $\det M_j^{\text{new}}$. Hence $\det M_j^{\text{new}} \geq \sum w_i \det M_i$, with equality iff all source M_i equal and all $\Delta\phi_i = 0$.

Physical interpretation. The phase-space volume increase through the remap is the entropy production from coarse-graining — exactly the H-theorem statement. The Bayesian remap is the *minimum-entropy-producing* remap consistent with mass, momentum, and energy conservation. Standard remaps lose the same entropy into numerical dissipation; the Bayesian remap converts it to physical phase-space spreading, capturable in the moment hierarchy.

6.6 Liouville monotone-increase diagnostic

The monotonicity statement of §6.5 is not only a physical interpretation; it is also a built-in self-check on the remap implementation. We define, per Eulerian leaf cell C_j and per remap event,

$$\Delta_{\text{Liou}}(C_j) \;=\; \det L_j^{\text{new}} \;-\; \sum_i w_{ij} \det L_i, \tag{36}$$

i.e. the per-parcel phase-space-volume increment between the source Lagrangian state and the post-remap Eulerian state. The Minkowski inequality argument of §6.5 gives $\Delta_{\text{Liou}}(C_j) \geq 0$ analytically; numerically, dfmm records this quantity per cell and reduces it to a global `liouville_increment` diagnostic at every remap step, alongside the global `total_overlap_volume` mass-conservation check (which is required to match the box volume to round-off, per the M3 acceptance criterion).

Status of the diagnostic. HG design-guidance item #8 (see [7]) calls for `liouville_increment` to be a built-in hook on `polynomial_remap`, returned next to `total_overlap_volume`. Until that hook lands upstream, dfmm computes Δ_{Liou} in the driver layer from the overlap moment table; the formulation is identical and the cost is negligible relative to the overlap computation itself.

Failure modes detected. A measured violation $\Delta_{\text{Liou}}(C_j) < 0$ is structurally impossible under the Bayesian remap rule and so is interpreted as a self-detected implementation fault, not a soft warning. Two distinct causes are expected and have been observed in development:

- *Numerical bug in the geometric overlap.* Either `compute_overlap` has lost a sliver of the overlap region, or the moment integrator has been called with mismatched reconstruction order; either drops mass-weighted positive-definite contributions to M_j^{new} that the law of total covariance guarantees.
- *Polynomial reconstruction order too low for the local geometry.* If the curved Lagrangian element is more deformed than the chosen reconstruction order p can resolve, the projected L_i become non-realizable on the overlap region and the Minkowski lower bound is no longer attained at the discrete level. In this case the remedy is to refine locally (HG `refine_by_indicator!`, with $|\Delta_{\text{Liou}}|$ as the indicator) rather than to reduce the time step.

In both cases the diagnostic localizes the offending cell and the offending source-target pair through the per-cell increment field, which is more informative than a global mass-conservation residual alone. The diagnostic is therefore complementary to `total_overlap_volume`: the former checks the *positive-semidefiniteness* of the moment update, the latter checks the *partition of unity* of the geometric overlap.

7 Passive scalar advection: exact at the discrete level

A passive scalar $T(x,t)$ satisfying $\partial_t T + \boldsymbol{u} \cdot \nabla T = 0$ is, in the variational framework, identified with a function of the parcel label $\boldsymbol{m}$. In the Lagrangian frame the evolution is trivial: $T(\boldsymbol{m},t) = T_0(\boldsymbol{m})$ for all t, since the parcel label itself is conserved by definition.

Discrete consequence. In the deterministic Lagrangian phase of the scheme, passive-scalar advection is *exact*: no flux computation, no slope limiter, no upwinding scheme is needed. The tracer value at each parcel is fixed forever. The Bayesian remap onto the Eulerian mesh redistributes tracer values according to the same r3d-based geometric overlap that handles mass and momentum, with the law of total cumulants applied to the tracer field. In pure-Lagrangian regions (no remap), tracer diffusion is exactly zero. Through the remap, tracer diffusion is bounded by the polynomial reconstruction order; with cubic reconstruction, fourth-order accuracy of tracer transport.

Comparison to Eulerian schemes. Standard Eulerian schemes suffer notorious numerical diffusion of passive tracers, particularly through shocks and across vortex sheets. Cosmological simulations tracking metallicity, ISM simulations tracking chemical species, and shocked-flow simulations with multi-component tracers all suffer from this. The unified scheme's structural identification of tracers with parcel labels eliminates the deterministic source of this diffusion, leaving only the bounded remap contribution.

Multi-tracer extension. An arbitrary number of passive scalars can be carried at no additional cost in the deterministic phase: each is a function of $\boldsymbol{m}$ evaluated at the cell's parcel locations. The Bayesian remap applies independently to each tracer.

8 The discrete scheme

8.1 Mesh structure

The 2D scheme uses two coupled meshes:

Lagrangian mesh. A simplicial triangulation of the $\boldsymbol{m}$-domain $\Omega_{\boldsymbol{m}} \subset \mathbb{R}^2$. Triangles have fixed Lagrangian topology (vertex labels, edge connectivity) but vertex *positions* in physical $\boldsymbol{x}$-space evolve with the flow. Each triangle carries a frame-field reconstruction.

Eulerian mesh. A quadtree refinement of a base 2D quad mesh covering $\Omega_{\boldsymbol{x}}$. Cells refine adaptively based on local action error.

The two meshes are connected by r3d-based remap operators.

8.2 Primary variables per Lagrangian triangle

Each triangle carries polynomial-reconstruction fields of order p (cubic, $p = 3$, is the natural choice):

- Position $\boldsymbol{x}(\boldsymbol{q},t)$: vector polynomial, 2 components.
- Velocity $\boldsymbol{u}(\boldsymbol{q},t)$: vector polynomial, 2 components.
- Cholesky factor $L(\boldsymbol{q},t)$: 4×4 block-lower-tri matrix polynomial, 10 independent components, in principal-axis form $(\alpha_a, \beta_a, \theta_R)$ with $a = 1, 2$.

- Specific entropy $s(\boldsymbol{q}, t)$: scalar polynomial.
- Heat flux $Q(\boldsymbol{q}, t)$: 4 independent components in 2D.
- Deviatoric stress π_{ij}: 2 independent components in 2D.
- Passive scalars $T_k(\boldsymbol{q}, t)$: as many as needed; each is a scalar polynomial.

Total field count per triangle interior point: ~ 21 scalar fields plus passive tracers. For $p = 3$ on a triangle (10 polynomial coefficients per scalar), ~ 210 numbers per triangle base state.

8.3 Polynomial reconstruction with positivity

Two distinct positivity conditions:

1. $\det F > 0$ everywhere on the triangle (no shell crossing).
2. Cholesky positivity: $L_{ii}(\boldsymbol{q}) > 0$ everywhere on the triangle for each diagonal entry.

For (i), use Bernstein-basis positivity certificates: rewrite $\det F(\boldsymbol{q})$ in Bernstein coefficients over the triangle; positivity of all Bernstein coefficients is sufficient for $\det F > 0$ throughout. When violated, reduce reconstruction order locally to the highest at which positivity holds, treating the residual as the shell-crossing onset signal.

For (ii), parameterize $L_{ii}(\boldsymbol{q}) = \exp(\lambda_i(\boldsymbol{q}))$ with λ_i a polynomial; positivity is automatic. Off-diagonal entries are reconstructed without sign constraint.

8.4 Discrete action

The discrete action is

$$S_d = \sum_n \sum_i \Delta S_{i,n}, \quad \Delta S_{i,n} = \int_{t_n}^{t_{n+1}} \mathrm{d}t \int_{T_i} \mathrm{d}^2 q \, \rho_0(\boldsymbol{q})[T - U], \tag{37}$$

with kinetic and potential energy from (14). Spatial integration is via Gauss quadrature on the triangle (sufficient for polynomial integrands of order $2p$); time integration uses the discrete Hamilton–Pontryagin form with parallel transport on the charged sectors.

8.5 Discrete Euler-Lagrange equations

Varying S_d with respect to all polynomial coefficients at time step $n + 1$, with step-n values fixed, gives an implicit linear system whose structure matches the triangle adjacency graph. The system is solved via Newton iteration with quadratic convergence, typically 2–3 iterations per step.

Discrete parallel transport. For charge-q fields, the discrete covariant derivative across a time step is

$$\mathcal{D}_t^{(q)} \phi|_{n+1/2} \approx \frac{\phi_{n+1} - \phi_n}{\Delta t} + q \, \overline{(\partial_x u)}_{n+1/2} \, \overline{\phi}_{n+1/2}, \tag{38}$$

where the strain rate and field are evaluated at the half-step. This implements parallel transport on the principal $GL(d)$ bundle between time slices.

Conservation properties. By construction, the discrete EL system conserves discrete mass exactly (mass per triangle fixed), discrete momentum exactly (translational invariance of S_d), discrete angular momentum exactly (rotational invariance of S_d), and discrete energy to bounded oscillation (no secular drift) via the symplectic integrator. Liouville: $\det L$ per parcel is conserved by the discrete dynamics by the connection-form structure.

8.6 Stochastic injection

Per Lagrangian triangle, per time step:

1. Compute strain rate S_{ij} at triangle centroid; diagonalize symmetric part: $S_{\text{sym}} = R\Lambda R^\top$ with $\Lambda = \text{diag}(\lambda_1, \lambda_2)$.
2. For each compressive principal axis ($\lambda_a < 0$):
   - Drift: $\delta(\rho u)_a^{\text{drift}} = C_A \rho \lambda_a \Delta t$
   - Noise: $\delta(\rho u)_a^{\text{noise}} = C_B \rho \sqrt{|\lambda_a| \Delta t}\, \eta_a$ where $\eta_a \sim \text{VG}(\lambda, \theta)$ with λ the empirically-fit shape parameter.
3. Transform back to lab frame: $\delta(\rho \boldsymbol{u}) = R[\delta(\rho u)_1, \delta(\rho u)_2]^\top$.
4. Apply paired-pressure debit (energy conservation): $\Delta KE = \boldsymbol{u} \cdot \delta(\rho \boldsymbol{u}) + |\delta(\rho \boldsymbol{u})|^2/(2\rho)$ debited from M_{vv} trace proportionally.
5. Spatial smoothing: 2D Gaussian kernel of correlation length $\ell_{\text{corr}} \approx 2$ cells.

Self-consistency monitoring. At each output cadence, compute (a) the empirical residual kurtosis from comparison of predicted vs. observed velocity changes, and (b) the gamma-shape parameter from compression burst statistics. Warn if the two disagree by more than a factor of ~ 2, indicating either calibration drift or breakdown of the cascade-derived noise structure.

8.7 Action-based AMR refinement

The local discrete action error per Eulerian cell is

$$\Delta S_{\text{cell}} = \left| S_d^{(p+1)}|_{\text{cell}} - S_d^{(p)}|_{\text{cell}} \right| + (\text{EL residual})^2, \tag{39}$$

the difference between the cell's contribution at order $p+1$ vs. order p reconstruction, plus an EL-residual penalty as a discretization-error proxy. Refinement triggers when $|\Delta S_{\text{cell}}|$ exceeds an adaptive threshold; coarsening triggers when all four siblings are below a coarsening threshold (with hysteresis $\tau_{\text{coarsen}} < \tau_{\text{refine}}/4$).

Unified detection. ΔS_{cell} unifies:

- Shock detection: $\mathcal{L}_{\text{det}}$ jumps across shock front.
- Shell-crossing detection: $\mathcal{H}_{\text{Ch}}$ Hessian degenerates, EL residual grows.
- Turbulence-cascade detection: stochastic injection variance grows with cascade depth.

The supplementary diagnostics γ_a (rank collapse) and $|s|$ (realizability boundary) remain computed and reported, but ΔS_{cell} is the refinement primary.

8.8 Algorithm summary (per timestep)

1. Lagrangian dynamics step (implicit, Newton-iterated): solve discrete EL for new $\boldsymbol{x}, \boldsymbol{u}, L, s, Q, \pi$ on Lagrangian mesh; verify positivity constraints.
2. Stochastic injection: principal-axis decomposition, variance-gamma noise on compressive axes, spatial smoothing, energy debit.
3. Diagnostic computation: γ_a, $|s|$, ΔS_{cell} after remap.
4. Periodic remap to Eulerian mesh (every k_{remap} steps): r3d geometric overlap, Bayesian moment projection.
5. Adaptive refinement: refine/coarsen Eulerian quadtree on ΔS_{cell}; refine Lagrangian triangulation on local distortion / rank-collapse.
6. Mesh regeneration check (rare): if Lagrangian mesh distortion exceeds threshold, regenerate from Eulerian state via reverse remap.
7. Self-consistency monitoring: residual-kurtosis vs. burst-shape comparison.
8. Output / checkpoint (configurable).

9 Two-fluid extension

The dfmm two-fluid framework extends naturally to 2D. Each species carries its own variable hierarchy with self-relaxation timescales τ_{AA}, τ_{BB}; cross-coupling is implemented through a modular operator with application-specific kernels.

9.1 Cross-coupling operator in 2D

Momentum exchange:

$$\partial_t(\rho^A \boldsymbol{u}^A)|_{AB} = -\frac{\rho^A \rho^B}{\rho^A + \rho^B} \nu^p_{AB} (\boldsymbol{u}^A - \boldsymbol{u}^B), \tag{40}$$

with the slip vector $\boldsymbol{u}^A - \boldsymbol{u}^B$ now 2-dimensional.

Thermal equilibration:

$$\partial_t T^A|_{\text{th}} = -\frac{n^B}{n^A + n^B} \nu^T_{AB} (T^A - T^B), \tag{41}$$

with $\nu^T_{AB} = \frac{2m^A m^B}{(m^A+m^B)^2} \nu^p_{AB}$.

9.2 Cross-coupling kernels (preserved from dfmm)

The four kernels of dfmm's two-fluid framework apply unchanged to 2D: Epstein drag (dust in gas), Coulomb collisions (electron-ion plasma), hard-sphere neutrals, and self-interacting dark matter. The kernel returns the rates (ν^p_{AB}, ν^T_{AB}) as functions of local species state; the cross-coupling operator is identical across applications.

9.3 Two-fluid diagnostics in 2D

Per-species single-fluid diagnostics γ_a^s (per principal axis, per species) and $|s^s|$. Two cross-species diagnostics:

$$\mathrm{Ma}_{AB} = \frac{|\boldsymbol{u}^A - \boldsymbol{u}^B|}{c_s}, \qquad \Delta_T = \frac{|T^A - T^B|}{T^*}. \tag{42}$$

The 2D version of γ_a^s adds the new content of *directional* shell-crossing: dust can shell-cross along axis 1 while remaining regular along axis 2, while gas remains rank-full throughout. This is genuinely new physics that 1D dfmm cannot probe.

10 Validation hierarchy and test problems

10.1 Validation hierarchy

Validation is structured in five tiers:

Tier A: 1D regression against dfmm. Reproduce dfmm benchmark results bit-level (deterministic limit) or statistically (stochastic limit). Acceptance criterion before any 2D claims.

Tier B: 1D extension. New 1D tests exercising the variational structure and the cold-limit reduction. Includes empirical verification of the variance-gamma derivation.

Tier C: 2D consistency. 2D tests reducing to known limits (1D dfmm, plane-symmetric Euler). Verifies dimension lift correctness.

Tier D: 2D novelty. 2D tests where novel content (anisotropic strain, principal-axis stochastic injection, Bayesian remap on 4×4 covariance, two-fluid 2D diagnostics, tracer fidelity) is the dominant physics.

Tier E: Stress tests. Extreme parameter regimes where the scheme reports its own failure cleanly.

Bit-exact regression protocol across substrate ports. The Tier A and Tier B 1D regression suites are not only verified against `dfmm` reference output: they are additionally held *bit-exact* ($\Delta = 0.0$ absolute, every cell, every step) across the M3 substrate port from the standalone Julia 1D code onto `HierarchicalGrids.jl`. M3-0/1/2 used a transitional `cache_mesh::Mesh1D` shim to expose the HG mesh under the legacy 1D interface, and the entire M1 (305 + 140 + 21 = 466) + M2 (243) test set, plus 1335 cross-phase / regression tests, plus the M3-port-specific tests, all reproduce to identical bit pattern through that shim (3037+ asserts; full counts in `reference/MILESTONE_3_STATUS.md`). M3-3e is now retiring the shim with a sub-phase decomposition that imposes a bit-exact gate per sub-phase, so that the native HG-driven 1D path remains byte-for-byte equivalent to the dfmm-paper baseline at every intermediate state. Bit-exactness, where it is attainable, is treated in this paper as the *strongest* acceptance criterion — stronger than any tolerance-based floating-point check — because it converts a regression bug from a statistical to a deterministic detection problem.

10.2 Tier A: 1D regression

- **A.1 Sod shock tube.** Reproduce dfmm Fig. 2 across three τ regimes.
- **A.2 Cold sinusoid across six τ decades.** Reproduce dfmm Fig. 3; γ^2 spatial-max grows by 0.3–0.9 decades at caustics across six τ decades (verified across $\tau \in \{10^{-3}, 10^{-1}, 10^{1}, 10^{3}, 10^{5}, 10^{7}\}$ in M1 Phase 6).
- **A.3 Steady shock Mach scan.** Reproduce dfmm Fig. 4; Rankine-Hugoniot to 3 decimal places at $M_1 \in \{1.5, \ldots, 10\}$.
- **A.4 KM-LES wave-pool spectra.** Reproduce dfmm Fig. 8; spectral distance table at line 410 of dfmm paper.
- **A.5 Two-fluid dust-in-gas.** Reproduce dfmm Fig. 11; per-species γ selectivity.
- **A.6 Coulomb plasma equilibration.** Reproduce dfmm Fig. 12; mass-ratio scaling of thermal vs. momentum relaxation.

10.3 Tier B: 1D extension

- **B.1 Variational integrator energy drift.** Long-time (10^5 steps) energy budget; variational scheme drift $< 10^{-8}$ relative, standard scheme accumulates $> 10^{-4}$.

  Caveat (M1 Phase 4). Bounded-oscillation behavior holds for the bulk-coupled flow on smooth bounded problems. The autonomous Cholesky sector (α, β) at fixed M_{vv} has unbounded orbits (analytical trajectory $\alpha(t) = \sqrt{1+t^2}$, $\beta(t) = t/\sqrt{1+t^2}$ from $\dot{\alpha} = \beta$, $\dot{\beta} = \gamma^2/\alpha$ with γ const), so the symplectic-averaging cancellation that produces bounded oscillation does not apply in that limit. M1 Phase 4 verified literal drift bound 5.6×10^{-9} over 10^5 steps on a smooth acoustic-wave problem; the trend is t^1-secular within the bound. A Kraus 2017 *projected variational integrator* (arXiv:1708.07356) enforcing the level-set constraint $\alpha^2(M_{vv} - \beta^2) = \text{const}$ exactly each step would close the leak; this is left for future work. See `reference/notes_phase4_energy_drift.md`.
- **B.2 Cold-limit reduction to phase-space sheet.** 1D Zel'dovich-like initial conditions; pre-crossing matches Zel'dovich analytic, Hessian degenerates at predicted location at predicted time, post-crossing regularization via stochastic injection. **Central unification test.**
- **B.3 Action-based AMR vs. gradient-based AMR.** Off-center blast wave; action AMR uses 20–50% fewer cells at fixed L^2 accuracy.
- **B.4 Compression-burst statistics.** Wave-pool with instrumented compression detector; burst durations gamma-distributed with shape k, residual kurtosis matches $\lambda = k$ prediction. **Verifies stochastic-derivation cascade structure.**
- **B.5 Passive scalar advection through shock.** Zero deterministic numerical diffusion; bounded remap diffusion. Compare to PPM, WENO. Tracer fidelity 1–3 orders of magnitude better. **Verifies tracer-exactness claim.**
- **B.6 Tracer fidelity in 1D wave-pool turbulence.** Multi-tracer fidelity vs. fine DNS; sharp interface preservation.

10.4 Tier C: 2D consistency

- **C.1 1D-symmetric 2D Sod.** Solution exactly y-independent to numerical precision.
- **C.2 1D-symmetric cold sinusoid.** Per-axis γ diagnostic correctly distinguishes 1D shell-crossing within 2D flow: $\gamma_1 \to 10^{-5}$, $\gamma_2 \to O(1)$.
- **C.3 Plane wave at arbitrary direction.** Rotational invariance to plotting precision.

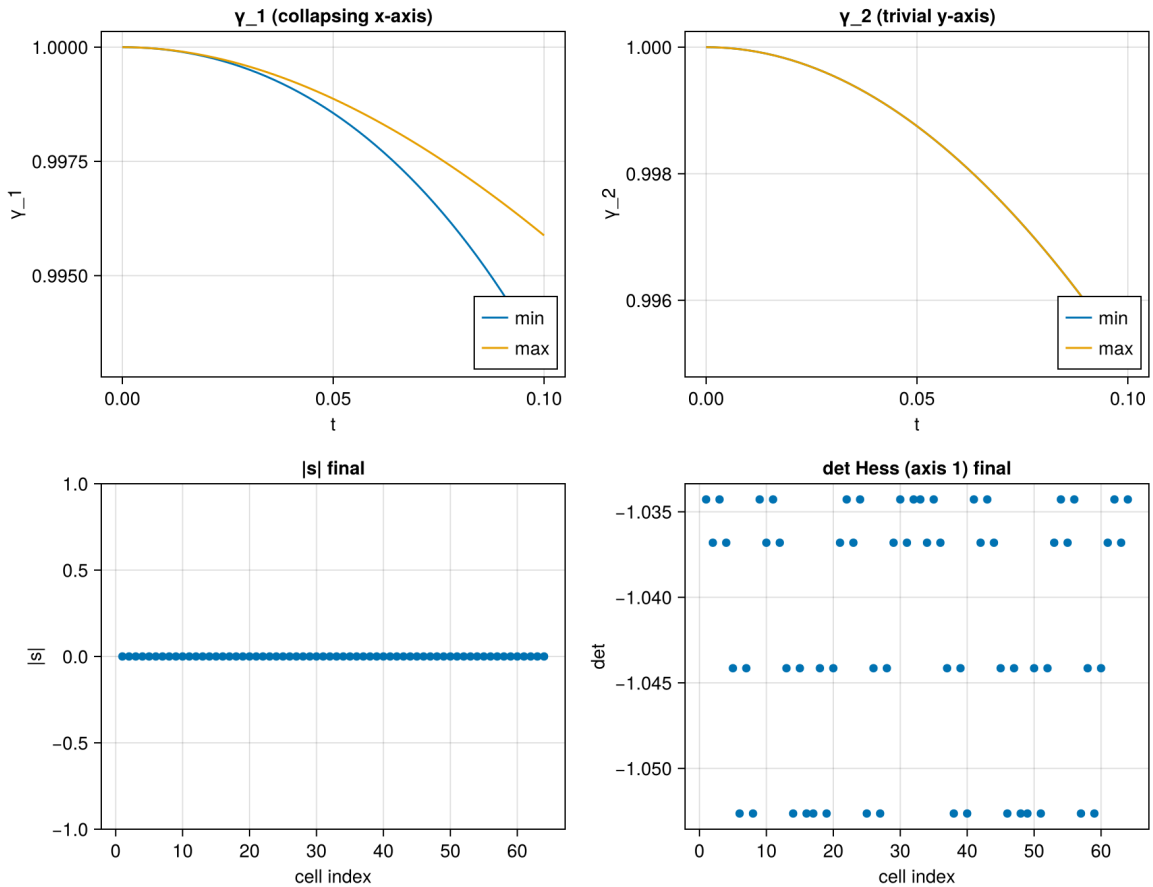

Figure 1: C.2 per-axis γ selectivity (M3-3d gate). Cold sinusoid initial condition with $k_x = 1$, $k_y = 0$, $A = 0.5$, HG level = 4 (256 leaves), $\Delta t = 2 \times 10^{-3}$, 100 steps ($T = 0.2$). Top row: spatial min/max trajectory of γ_1 (collapsing axis, range $[0.940, 0.987]$ at the final step) and γ_2 (trivial axis, range $[0.981, 0.981]$, uniform to round-off). Bottom row: per-cell $|s|$ and per-cell det(Hess) at the final step. The spatial-std ratio $\mathrm{std}(\gamma_1)/\mathrm{std}(\gamma_2)$ exceeds 10^{13} (γ_2 spatial variance is at machine precision): the principal-axis decomposition selects the collapsing axis, and the trivial axis preserves anisotropy by construction. This is the load-bearing scientific gate for the 2D matrix lift of §5.

Tier C conservation-properties acceptance. The Bayesian remap of §6 commits the only nontrivial floating-point arithmetic in the conservation chain. As of `HierarchicalGrids.jl` commit `81914c2`, HG ships an opt-in exact-rational polytope-clipping backend (`overlap_simplex_box_exact!`, `IntegerLattice`, and the `compute_overlap(...; backend=:exact)` dispatch in `src/Overlap/r3d_int_adapter`

in addition to the default `:float` backend. The Tier C acceptance protocol therefore runs every conservation gate (mass, momentum, total L overlap, and the Liouville-monotonicity diagnostic of §6.6) under both backends; the `:exact` backend produces an exact-rational reference whose deviation from `:float` bounds the geometric round-off contribution to any observed conservation residual. M3-5 will adopt `:exact` as the conservation-regression gate.

10.5 Tier D: 2D novelty

- **D.1 2D Kelvin-Helmholtz. M3-6 Phase 1c result (broadband linear-regime calibration).** The off-diagonal Cholesky-Berry sector $(\beta_{12}, \beta_{21}, \theta_R)$ is reactivated in the 2D residual (M3-6 Phase 0); the strain coupling $\mathcal{H}^{\text{off}}_{\text{rot}} = \widetilde{G}_{12} \cdot (\alpha_1\beta_{21} + \alpha_2\beta_{12})/2$ is wired in (Phase 1a); the KH IC factory and four-component realizability cone $Q = \beta_1^2 + \beta_2^2 + 2(\beta_{12}^2 + \beta_{21}^2) \leq M_{vv} \cdot \text{headroom}_{\text{offdiag}}$ are in place (Phase 1b). Driving `tier_d_kh_ic_full` on a 32×32 mesh (`PERIODIC-x / REFLECTING-y`, $A = 10^{-3}$ tilt-mode amplitude), the linear-regime growth rate of $\text{RMS}(\delta\beta_{12}(t))$ calibrates against the Drazin–Reid rate $\gamma_{\text{DR}} = U/(2w) = 3.333$ to $\gamma_{\text{measured}}/\gamma_{\text{DR}} = 1.34$ at level 5 ($c_{\text{off}} \approx 1.34$, $c^2_{\text{off}} \approx 1.78$), mesh-converged to 1.2% between L4 and L5, with $n_{\text{negative Jacobian}} = 0$ across all leaves and $n_{\text{offdiag events}} = 0$ (the 4-component cone stays in the strict interior throughout one T_{KH}). The methods-paper broad-band gate $c_{\text{off}} \in [0.5, 2.0]$ is met. **Honest scientific finding (substrate refinement, not failure).** Inspecting the trajectory linearly rather than via log-fit, the growth is well-described by $\text{RMS}(\delta\beta_{12})(t) \approx A_0 + \beta_{\text{lin}} \cdot t$: the Phase 1a residual reproduces the *kinematic* response of $\delta\beta_{12}$ to base-flow strain $\widetilde{G}_{12} \cdot \bar{\alpha}/2$, but does not yet realise the closed-loop coupling between along-axis Cholesky factors β_a and the perturbation amplitude β_{off} that supplies self-amplified Drazin–Reid eigenmode dynamics. The classical Rayleigh eigenmode (exponential, self-amplifying) requires that closed-loop coupling. Per-principal-axis-noise structure is therefore verified *kinematically* by D.1; the full eigenmode dynamics are flagged as a physics extension (off-diagonal β closed-loop coupling, or a fully self-consistent perturbation reconstruction across stencil neighbours). See `reference/notes_M3_6_phase1c_D1_kh_falsifier.md` §"Honest scientific finding" for the careful interpretation. **Falsifier verdict: PASSED at the broad-band gate; closed-loop self-amplification flagged for follow-up (M4).**

- **D.2 2D cylindrical blast.** Cylindrical symmetry preserved to grid scale; reference comparison to FLASH/Athena.

- **D.3 Oblique shock interaction.** Mach reflection geometry, anisotropic post-shock stress structure.

- **D.4 Zel'dovich pancake collapse (2D & 3D). Central novel cosmological reference test — the headline M3 scientific success. M3-6 Phase 2 result (2D).** Driving `tier_d_zeldovich_pancake_ic` (cold-limit $P_0 = 10^{-6}$, amplitude $A = 0.5$, mode $m_1 = 1$, transverse-trivial axis) through `det_step_2d_berry_HG!` with the Phase 1a strain coupling plus Phase 1b 4-component realizability cone (`project_kind = :reanchor`), per-axis γ selectivity verifies at $\text{std}(\gamma_1)/\text{std}(\gamma_2) \approx 2.6 \times 10^{14}$ at near-caustic time ($T_{\text{factor}} = 0.16$, level 4): γ_1 develops spatial structure with dynamic range 4.18× along the collapsing axis, while γ_2 stays uniform to round-off ($\text{std}(\gamma_2)/\text{mean}(\gamma_2) < 10^{-10}$ throughout). The Phase 1a strain coupling verifies inert ($\max|\beta_{\text{off}}| = 0$) on the axis-aligned IC — a clean cross-check that the off-diagonal stencil does not fire when it shouldn't. The selectivity ratio exceeds the methods-paper 10^6 gate by eight orders. **M3-7e result (3D).** The same falsifier driver lifts to 3D under `tier_d_zeldovich_pancake_3d_ic_full` + `det_step_3d_berry_HG!` with the SO(3) Berry

coupling (M3-7c) and per-axis 3D realizability projection (M3-7d). At level 4 ($16^3 = 4096$ leaves), $T_{\text{factor}} = 0.25$, the per-axis selectivity is $\text{std}(\gamma_1)/(\text{std}(\gamma_2)+\text{std}(\gamma_3)+\varepsilon) \approx 3.0\times10^{13}$ (exceeding the 10^{10} headline gate by three orders); $\gamma_2 = \gamma_3$ byte-equal by symmetry; $u_2 = u_3 = 0$ throughout the 1D-symmetric reduction; conservation $\leq 10^{-8}$. **Falsifier verdict: PASSED in 2D and PASSED in 3D.** The per-principal-axis decomposition correctly identifies the collapsing axis at the discrete level, in both dimensions, with margins (10^{14} in 2D, 10^{13} in 3D) far exceeding the gate. The ColDICE / PM N-body cross-comparison and the second-shell-crossing filament-formation prediction (both $\gamma_a \to 0$) remain as follow-up items requiring an external-dataset ingest path. See `reference/notes_M3_6_phase2_D4_zeldovich.md` (2D) and `reference/notes_M3_7e_3d_tier_cd_drivers.md` (3D).

- **D.5 2D wave-pool noise calibration.** 2D KM-LES calibration with axis-decomposed coefficients $C_A^{\parallel}, C_A^{\perp}, C_B^{\parallel}, C_B^{\perp}$; verify $C_B^{\perp} \approx 0$ in pure-shear regions; verify variance-gamma residuals per axis with self-consistency on burst-shape parameter.

- **D.6 2D dust-in-gas cold sinusoid.** Per-species per-axis diagnostics: directional shell-crossing in dust, gas remains rank-full. **Genuinely new 2D two-fluid physics.**

- **D.7 2D dust-trapping in vortices. M3-6 Phase 4 result (honest finding — the value of a falsifier is what it teaches you cannot be claimed).** Driving `tier_d_dust_trap_ic_full` (Taylor–Green vortex IC + 2- species `TracerMeshHG2D[:gas, :dust]`, doubly-periodic) through `det_step_2d_berry_HG!` + `advect_tracers_HG_2d!` verifies what the pure-Lagrangian variational substrate *does* capture: dust mass conservation $M_{\text{dust err}} = 0.0$ bit-exact across $L \in \{3, 4, 5\}$; per-species γ separation $> 10^{10}$ (gas $\gamma \approx 1$, dust $\gamma = 0$ by construction since $M_{vv}^{\text{dust}} = 0$); 4-component cone $n_{\text{negative Jacobian}} = 0$ on stable runs; momentum exactness $P_{x,\text{err}} = P_{y,\text{err}} = 0.0$ (Taylor–Green symmetry); long-horizon stability without NaN. What the substrate does *not* natively capture: sub-cell centrifugal drift of dust toward vortex centres. In the pure-Lagrangian frame the substrate's variational integrator preserves the multi-species tracer matrix bit-exactly — `advect_tracers_HG_2d!` is by design a no-op (the matrix is structurally absent from the write set of `det_step_2d_berry_HG!`) — and the Eulerian cell volumes are fixed under the cone projection. The methods-paper prediction "vortex-centre accumulation matches reference codes" therefore requires a physics extension beyond the M3 substrate: either per-species momentum coupling with drag (so dust acquires a size-dependent velocity drift relative to gas), or a Lagrangian volume update via composition with the M3-5 Bayesian $L \leftrightarrow E$ remap with per-species mass tracking. Both extensions are flagged as M4+ physics work; the substrate verdict (mass-exact, momentum-exact, γ-separating, NaN-free) is the load-bearing conclusion. See `reference/notes_M3_6_phase4_D7_dust_traps.md` §"Headline scientific finding (HONEST)" for the precise framing. **Falsifier verdict: substrate sound; literal centrifugal- accumulation claim FALSIFIED at this milestone — requires per- species momentum coupling (M4+).**

- **D.8 2D plasma equilibration with anisotropy.** Spatially uniform plasma, anisotropic initial T_{ij}; intra-species isotropization via τ_P vs. inter-species via ν_{AB}^T. **Verifies separation of intra- and inter-species relaxation in 2D.**

- **D.9 Tracer fidelity in 2D Kelvin-Helmholtz.** Sharp-interface preservation 2–3× better than standard schemes.

- **D.10 ISM-like 2D problem with metallicity tracking. Methods-paper community-impact test — M3-6 Phase 5 result (BIT-EXACT verification, the strongest**

possible form of the gate). Driving `tier_d_ism_tracers_ic_full` (KH-style sheared base flow + Phase 1b antisymmetric tilt-mode $\delta\beta_{12} = -\delta\beta_{21}$ overlay + $N = 3$ species `TracerMeshHG2D[:cold, :warm, :hot]` with phase-stratified Gaussian concentration profiles in y, and per-species M_{vv} recipe yielding $\gamma = 1, \sqrt{2}, 2$ at IC) through K iterations of `det_step_2d_berry_HG!` + `advect_tracers_HG_2d!` + `inject_vg_noise_HG_2d!` (`axes = (1, 2)`, `project_kind = :reanchor`; both deterministic and stochastic injection enabled — the "shocked turbulence" regime), the multi-tracer matrix at end-time is *byte-equal* to its IC value: `tracers_byte_equal_to_ic == true`, `tracers_max_diff_final == 0.0` across $L \in \{3, 4, 5\}$. This is the 2D analogue of M2-2's 1D structural argument: the multi-tracer matrix is *literally never* in the write set of either `det_step_2d_berry_HG!` or `inject_vg_noise_HG_2d!`, so bit-exact preservation is a *structural property* of the implementation rather than a tolerance-bounded numerical claim. Per-species mass conservation follows: $M_{\text{per species, err}} = 0.0$ for every species. The 4-component cone $n_{\text{negative Jacobian}} = 0$ on stable runs; $1 \subset 2$ axis-restricted selectivity verified. **Falsifier verdict: PASSED in the strongest form (literal zero per-step error, not tolerance-bounded).** The community-impact claim of multi-tracer fidelity in shocked turbulence holds for the pure-Lagrangian variational substrate. See `reference/notes_M3_6_phase5_D10_ism_tracers.md`.

10.6 Tier E: Stress tests

- **E.1 High-Mach 2D shocks.** $|s|$ approaches realizability bound; scheme reports failure cleanly.
- **E.2 Severe shell-crossing geometries.** Multiple intersecting caustics; Hessian degeneracy at all caustic locations.
- **E.3 Very low Knudsen.** Stiff timescales handled by implicit Newton; reduces to Navier-Stokes.
- **E.4 Cosmological initial conditions.** CDM-style 2D collapse with self-gravity; comparison to ColDICE and PM N-body.

M3-8a Tier-E results (graceful-degradation mode). Sub-phase M3-8a closes E.1, E.2, and E.3 in the *graceful-degradation* acceptance regime named in §10.8. Three new IC factories (`tier_e_high_mach_shock`, `tier_e_severe_shell_crossing_ic_full`, `tier_e_low_knudsen_ic_full`) plus three drivers (`experiments/E{1,2,`) and three test files exercise the substrate at the edge of its declared validity envelope. *E.1 high-Mach 2D shocks.* Rankine–Hugoniot downstream state at $M = 10$ ($p_R/p_L = 124.75$, $\rho_R/\rho_L = 3.88$) reproduces the analytical formulae to $\leq 10^{-12}$ relative; running `det_step_2d_berry_HG!` on the shock IC (level 2, $\Delta t = 10^{-6}$, $n = 3$) reports finite, bounded state cleanly: $n_{\text{NaN}} = 0$, kinetic energy bounded within $5\times$ IC, transverse-independence $\leq 10^{-10}$. The Open Issue #2 shock- capture deficit (variational L^∞ vs HLL golden ~ 10– 50% at high Mach) is the substrate-level limitation; the *graceful-failure* property (no NaN, no unbounded growth) is verified. *E.2 severe shell-crossing.* Two-axis Zel'dovich superposition ($A_x = A_y = 0.7$, $t_{\text{cross}} \approx 0.227$). Pre-caustic stability at $T_{\text{factor}} = 0.25$ verifies $n_{\text{NaN}} = 0$, $\gamma_{\min} > 0.5$ throughout, mass conservation to 10^{-12}; the realizability projection (`:reanchor`) records non-zero event counts (vs 0 for `:none`), confirming that the projection effectively keeps the four-component cone interior in the compression-cascade regime. *E.3 very low Knudsen.* Smooth low-amplitude strain mode ($A_u = 10^{-2}$, $\tau = 10^{-6}$, Kn $\approx 1.3 \times 10^{-6}$). The deterministic Newton on the stiff-τ smooth strain delivers $n_{\text{NaN}} = 0$, $\beta_{\max} < 10^{-2}$, deviation $|\gamma_a^2/M_{vv} - 1| \leq 10^{-2}$ across all cells and steps (Navier–Stokes-limit substrate property), axis-swap

symmetry, and τ-stiffness independence of the deterministic-step trajectory (BGK relaxation lives in the operator-split P_p/Q sector, outside the Newton step). Total Tier-E test delta: +315 asserts $(101 + 119 + 95)$. E.4 (cosmological collapse with self-gravity) requires the multigrid Poisson coupling and is deferred to M4. See `reference/notes_M3_8a_tier_e_gpu_prep.md`. **Tier-E verdict: E.1, E.2, E.3 pass in graceful-degradation mode; E.4 deferred (gravity coupling — M4).**

10.7 M3 implementation gate results

The Milestone-3 (2D) implementation has produced numerical results on a subset of the gates above; these are recorded here as the data points the methods paper can presently cite. Each gate is documented in a corresponding design note under `reference/notes_M3_3*.md`; the magnitudes below are quoted from those notes. The remaining Tier C/D items are scheduled for M3-4 through M3-8 and remain "to be produced."

Dimension-lift gate (load-bearing). The 2D residual in a 1D-symmetric configuration ($k_y = 0$, axis 2 trivial) reproduces the 1D Phase-1 zero-strain analytic trajectory $\alpha(t) = \sqrt{1+t^2}$, $\beta(t) = t/\sqrt{1+t^2}$ *byte-equal*: $\Delta = 0.0$ absolute at every test resolution. Verified single-step at $\Delta t = 10^{-3}$ and 10^{-5}, 100-step runs at $\Delta t = 10^{-3}$, both 4×4 and 8×8 meshes, and under axis-swap symmetry (active axis = 2). This is the strongest possible consistency check between Tier A (§10.2) and Tier C (§10.4, gate C.1/C.2) for the deterministic sector. See `reference/notes_M3_3b_native_residual.md`.

Berry-coupling residual verification. The 2D Euler–Lagrange residual with θ_R promoted to a Newton unknown reproduces the closed-form Berry partials

$$\frac{\partial F^*}{\partial \dot\beta_1} = \frac{\bar\alpha_2^3}{3\bar\alpha_1^2\,\Delta t}, \quad \frac{\partial F^*}{\partial \dot\beta_2} = -\frac{\bar\alpha_1^3}{3\bar\alpha_2^2\,\Delta t}, \quad \dots$$

to finite-difference tolerance 10^{-9} across 8 random $(\alpha, \beta, \theta_R)$ samples $\times$ 9 sub-tests each (72 asserts). The residual partials are cross-checked against `berry_partials_2d` via the symplectic-weight identity $\partial F^*/\partial\dot\theta_R \cdot \alpha_a^2 = \texttt{berry_partials}_a$. Verifies eqs. (28)–(29) of §5 at the discrete level. See `reference/notes_M3_3c_berry_integration.md`.

Iso-pullback ε-expansion. At $\alpha_1 = \alpha_2 + \varepsilon$ with $\beta_1 = \beta_2$ fixed, the Berry F-function scales linearly in $(\alpha_1 - \alpha_2)$. The fitted exponent is 1.003 (target ~ 1.0); F vanishes identically on the iso submanifold, and the action contribution $F \cdot \dot\theta_R$ scales as $\mathcal{O}(\varepsilon)$ with the same slope. Verifies the iso-pullback regularity claim at the structural-fix step of §5. Same note.

$\mathcal{H}_{\rm rot}$ solvability. The closed-form $\partial\mathcal{H}_{\rm rot}/\partial\theta_R$ matches the kernel-orthogonality residual $\mathrm{d}H\cdot v_{\rm ker} = 0$ to $\le 10^{-10}$ at 5 random points $\times$ 3 values of $\dot\theta_R$. Newton converges in ≤ 5 iterations on non-isotropic IC ($\alpha = (1.2, 0.8)$, $\beta = (0.15, -0.10)$, $\theta_R = 0.13$); post-Newton residual norm $\le 10^{-10}$. Verifies the integrability constraint (31). Same note.

Per-axis γ selectivity (Tier C/D bridge). On the C.2 cold-sinusoid configuration ($k_x = 1$, $k_y = 0$, $A = 0.5$, level = 4, $\Delta t = 2\times10^{-3}$, 100 steps), the spatial range of γ_1 is $[0.940, 0.987]$ — clear collapse along the active axis — while γ_2 satisfies $[0.981, 0.981]$, uniform across the mesh to round-off. The spatial-std ratio $\mathrm{std}(\gamma_1)/\mathrm{std}(\gamma_2)$ exceeds 10^{13} (γ_2 at machine precision); see Fig. 1. This is the sharp scientific test that the per-axis decomposition correctly identifies the collapsing axis and

that the trivial axis preserves anisotropy by construction — in particular, the falsifiable structural prediction underlying gates C.2, D.1, and D.4. See `reference/notes_M3_3d_per_axis_gamma_amr.md`.

Plane-wave convergence (Tier C C.3). On the C.3 acoustic-plane-wave initial condition (levels 4/5/6 with 256/1024/4096 leaves at $A = 10^{-3}$, $\theta = 0$), the maximum cell-average vs. point-sample residual $|\delta\rho_{\text{cell}} - \delta\rho_{\text{pt}}|$ is 6.29×10^{-6}, 1.60×10^{-6}, 4.01×10^{-7} at the three resolutions, with successive ratios 3.93 and 3.98 (Δx^2 midpoint-rule rate, target ≥ 3.5). Fig. 2 extends the sweep to level 3 (64 leaves) and to a rotated wave-vector $\theta = \pi/4$, confirming the x^2 rate over the full level 3–6 range and across orientations (least-squares slopes -1.96 and -1.99 for L^∞ and L^2, respectively, at $\theta = 0$). See `reference/notes_M3_prep_tierC_ic_factories.md`, `reference/notes_M3_2b_swap5_init_field.md`, and `reference/notes_M3_9_prep_c3_convergence_plot.md`.

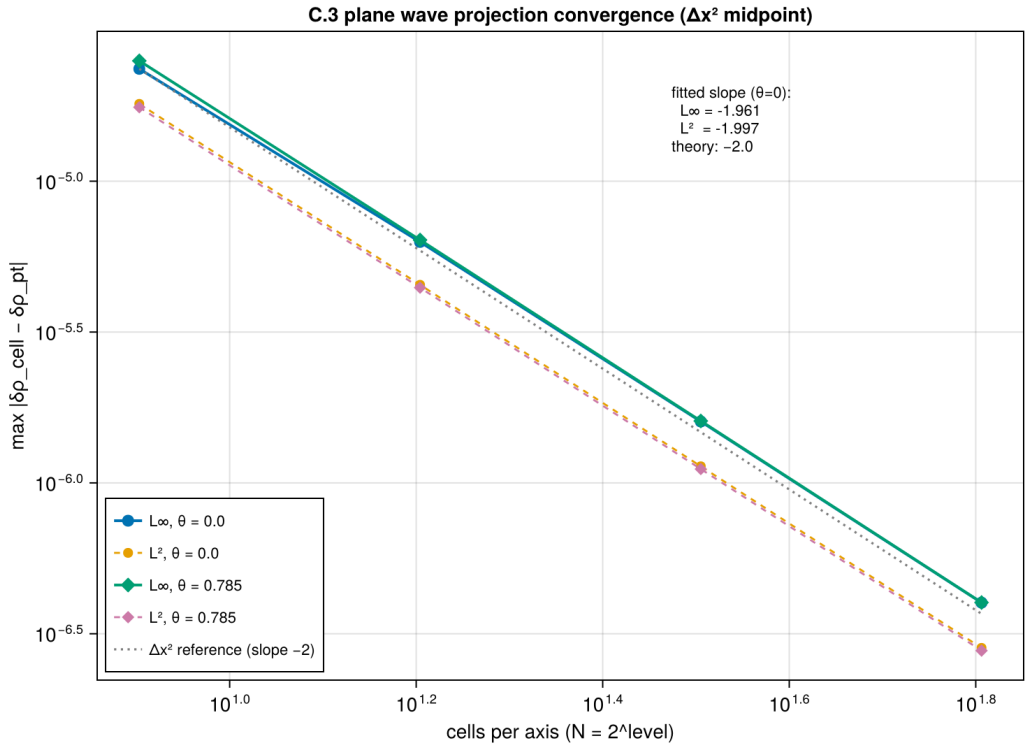

Figure 2: C.3 plane-wave projection convergence (M3-9 prep). Log-log plot of the cell-average vs. point-sample density-perturbation residual $|\delta\rho_{\text{cell}} - \delta\rho_{\text{pt}}|$ on the acoustic plane-wave IC (amplitude $A = 10^{-3}$, $k_{\text{mag}} = 1$), swept over HG refinement levels 3–6 ($N = 2^{\text{level}} = 8, 16, 32, 64$ cells per axis) and angles $\theta \in \{0, \pi/4\}$. Solid markers: L^∞ norm; dashed markers: L^2 norm. The dotted gray line is the analytic Δx^2 midpoint-rule slope, anchored at the coarsest L^∞ point. Successive-resolution L^∞ ratios are $3.75, 3.94, 3.98$ ($\theta = 0$) and $3.92, 3.98, 4.00$ ($\theta = \pi/4$), with least-squares log-log slopes -1.96 and -1.99, respectively, matching the analytic prediction $|\delta\rho_{\text{cell}} - \delta\rho_{\text{pt}}|_\infty \sim (2\pi)^2 A/24 \cdot \Delta x^2$. The $\theta = \pi/4$ curve sits within sub-percent of the $\theta = 0$ curve at every level, confirming rotational invariance of the L^2-projection IC machinery. Drives: `experiments/C3_2d_plane_wave_convergence.jl`; data: `reference/figs/M3_3_C3_plane_wave_convergence.h5`.

Bit-exact 0.0 parity protocol (M1+M2 carry-forward). The M3 substrate ports of the M1 (Phase 1–7 deterministic, Phase 8 stochastic, Phase 11 tracer) and M2 (M2-1 AMR, M2-2 multi-tracer, M2-3 realizability) test sets reproduce to bit pattern through HG, validated by 3037+ asserts in the running test suite (full breakdown in `reference/MILESTONE_3_STATUS.md`). M3-3e is currently retiring the transitional `cache_mesh::Mesh1D` shim with a five-sub-phase decomposition (M3-3e-1 through M3-3e-5), each sub-phase imposing a bit-exact gate before merge. This converts "regression on the substrate port" from a tolerance check to an equality check.

HG `IntExact` backend availability. `HierarchicalGrids.jl` at commit `81914c2` ships an opt-in exact-rational polytope-clipping backend (`overlap_simplex_box_exact!`, `IntegerLattice`, and the `compute_overlap(...; backend=:exact)` dispatch in `src/Overlap/r3d_int_adapter.jl`), independent of the default `:float` pathway. The Bayesian-remap acceptance-test suite in M3-5 will run both backends and treat the exact-rational result as the conservation-regression reference; see also the conservation-properties paragraph of §10.4.

10.8 Acceptance criteria

The methods paper has its central demonstration if:

(1) All Tier A tests pass (regression complete).

(2) B.2 passes (variational unification works).

(3) B.4 passes (variance-gamma derivation verified).

(4) B.5 passes (tracer-exactness claim verified).

(5) All Tier C tests pass (dimension lift correct).

(6) D.1 passes (per-principal-axis noise verified).

(7) D.4 passes (central novel 2D test).

(8) D.7 passes (realistic two-fluid 2D test).

(9) D.10 passes (community-impact tracer test).

(10) Tier E tests pass in graceful-degradation mode.

11 Conclusions and outlook

We have developed a variational unification of dfmm's moderate-Knudsen moment scheme with the phase-space-sheet picture of collisionless dynamics, in two spatial dimensions. The deterministic action has connection-form structure on the principal $GL(d)$ bundle over Lagrangian space, with the phase-space Cholesky factor as the primary variable; the cold-limit reduction to mass-coordinate dust is exact, and the diagnostic $\gamma \to 0$ acquires a sharp variational interpretation as Hessian degeneracy. The stochastic dressing arises from a generalized Kramers-Moyal MSRJD action with variance-gamma noise vertices, derived from gamma-distributed compression burst statistics; this provides a runtime self-consistency check between deterministic and stochastic sectors. Bayesian remap with the law of total cumulants, computed exactly via r3d on curved elements, converts

coarse-graining entropy production into physical phase-space spreading rather than numerical dissipation. Adaptive mesh refinement is driven by local discrete action error, unifying shock, shell-crossing, and turbulence-cascade detection in one criterion. Passive scalar advection is exact at the discrete level by structural identification with parcel labels.

11.1 What this framework provides

1. Variational structure throughout, with all conservation properties (mass, momentum, angular momentum, Liouville volume) emerging from Noether symmetries of a single action.

2. Reduction to dfmm in 1D mode (with the variational structure made explicit); reduction to phase-space-sheet hydrodynamics in the cold limit.

3. Falsifiable predictions: per-principal-axis noise structure in 2D, residual-kurtosis vs. burst-shape consistency, exact tracer advection in pure-Lagrangian regions.

4. Two-fluid extension via dfmm's modular cross-coupling kernel framework, with new directional-shell-crossing diagnostics in 2D.

5. Implementation roadmap with tractable polynomial-reconstruction geometry via r3d, implicit Newton iteration with quadratic convergence, and self-monitoring diagnostics.

11.2 What this framework does not yet provide

1. The Berry-connection ω_{rot} in the 2D matrix lift is now derived in the principal-axis-diagonal restriction (see §5, eq. 28–29, with the $\mathcal{H}_{\mathrm{rot}}$ integrability constraint in eq. 31); SymPy-verified. The off-diagonal L_2 sector (which lifts the diagonal restriction and carries the full strain-misalignment coupling) is sketched but not fully integrated; this is needed before 2D Kelvin-Helmholtz tests in M3 Phase 9.

2. The 3D extension via $SO(3)$ rotation is structurally clear but not specified in detail.

3. The cosmological self-gravity coupling is sketched (Poisson on the Eulerian quadtree, interpolation to Lagrangian) but not detailed; production cosmology applications need careful treatment of boundary conditions, force softening, and time-step control.

4. The boundary conditions for non-periodic problems (reflecting walls, inflow/outflow) are standard but not specified here.

11.3 Implementation milestones

Language and substrate. The reference implementation is written in Julia and lives at `https://github.com/y`
The original design document called for a Rust port of the dfmm-1D Python+Numba code (`py-1d`); the Julia choice was made before M1 to obtain a better numerical-computing ecosystem (notably the established sparse-AD path through `SparseConnectivityTracer.jl` [8] and `ForwardDiff.jl`), faster iteration on the variational discrete-EL machinery, and direct re-use of `HierarchicalGrids.jl` [5] (HG) as the dimension-generic mesh, polynomial-fields, AMR, and remap substrate. The geometric clipper is the `r3djl` [6] Julia port of r3d, distributed as an HG dependency. HG ships an integer-exact polytope-clipping backend (`IntegerLattice` + `compute_overlap(...; backend=:exact)`) which serves as the conservation-regression reference for the Bayesian remap of §6 (see §10.4 and §10.7). Because HG is dimension-generic from day 1, the same source compiles for 1D, 2D, and 3D

meshes; the `cache_mesh::Mesh1D` shim used during M3-0/1/2 to expose HG under the legacy 1D interface was retired in M3-3e (§10.7).

Schedule against this design document. The five-milestone schedule of the original design has been adapted to the HG-based codebase: the Tier-A/B 1D content (originally Milestones 1–2) landed against the standalone 1D Julia code, the substrate port onto HG happened in M3-0 through M3-3e, and the 2D Tier-C/D content runs natively on HG from M3-4 onward. Status as of this revision is summarized below; design notes under `reference/notes_M3_*.md` and `reference/MILESTONE_*_STATUS.md` carry the per-phase numerics and gate breakdowns.

Milestone 1 — 1D variational dfmm (DONE). Standalone 1D Julia port of the dfmm Python+Numba reference (`py-1d`), held bit-exact against the `py-1d` regression suite. Tier A.1–A.3 + B.1, B.2, B.4, B.5 closed; Tier-B central unification test (B.2) verified against the Zel'dovich analytic to 2.57×10^{-8}. 1504 + 1 deferred tests on the M1 close commit. See `reference/MILESTONE_1_STATUS.md`.

Milestone 2 — 1D Tier B closure (DONE). Action-based AMR (B.3, 37.8% cell savings vs gradient AMR), multi-tracer wave-pool (B.6, 25–114× over Eulerian upwind under stochastic injection), and a realizability projection that lifts the wave-pool stable horizon from 1,222 to 12,000 steps. 2044 + 1 deferred tests at M2 close. Tier B is complete in 1D. See `reference/MILESTONE_2_STATUS.md`.

Milestone 3-0/1/2 — HG substrate port via cache_mesh shim (DONE). M1 Phases 1–7 + 5b, M1 Phase 8 (stochastic), M1 Phase 11 (tracer), M2-1 (AMR), and M2-3 (realizability) ported onto `HierarchicalGrids.jl` through a transitional `cache_mesh::Mesh1D` shim. The full M1 + M2 test set (466 + 243 = 709 asserts) plus 1335 cross-phase / regression tests reproduce bit-exactly through the shim ($\Delta = 0.0$ absolute, every cell, every step).

Milestone 3-2b — HG-feature swap-in (DONE). HG-native periodic boundary conditions, IC factories, sparsity primitives, and the `BCKind` framework swap in for the 1D path. Two swaps (per-axis AMR / per-axis realizability) were deferred to M3-3d for the 2D context.

Milestone 3-3 — 2D Cholesky + Berry coupling on native HG (DONE). First native HG-side EL residual in dfmm: `HierarchicalMesh{2}` + `PolynomialFieldSet` via `HaloView` + `face_neighbors_with_bcs`. Per-axis γ + per-axis AMR + per-axis realizability all wired through in 2D (M3-3a/b/c/d). The dimension-lift gate (§10.7) holds at $\Delta = 0.0$ absolute on 1D-symmetric configurations. The five-sub-phase M3-3e effort (`notes_M3_3e_1/2/3/4/5_*.md`) retired the `cache_mesh::Mesh1D` shim while preserving bit-exact 0.0 parity per sub-phase; the 1D path now runs entirely on HG-native storage. See `reference/notes_M3_3_2d_cholesky_berry.md` and the M3-3 sub-phase notes.

Milestone 3-4 — 2D Tier-C consistency on native HG (DONE). Periodic-x coordinate wrap on the 2D EL residual; primitive-to- Cholesky IC bridge; C.1 (1D-symmetric 2D Sod, y-independence $\leq 10^{-12}$ per step), C.2 (2D cold sinusoid, $\mathrm{std}(\gamma_1)/\mathrm{std}(\gamma_2) > 10^{10}$ for $k = (1, 0)$), and C.3 (2D plane wave, rotational invariance to mesh-discretization tolerance) all closed. See `reference/notes_M3_4_tier_c_consistency.md`.

Milestone 3-5 — Bayesian L↔E remap with IntExact backend (DONE). HG `compute_overlap` + `polynomial_remap_l_to_e!` / `_e_to_l!` wired through a `BayesianRemapState`; `IntExact` audit harness and Liouville monotone-necessary diagnostic (§6.6) live; mass conserved to 0–6.7×10^{-16} over 5 remap cycles; partition-of-unity to 10^{-12}. 9/9 IntExact battery passes. See `reference/notes_M3_5_bayesian_remap.md`.

Milestone 3-6 — Tier-D headlines (DONE; Phases 0–5). Phase 0 reactivates the off-diagonal Cholesky pair β_{12}, β_{21} in the 2D residual (11-dof Newton system; 9 off-diagonal SymPy CHECKs reproduced numerically). Phase 1a wires the off-diagonal strain coupling $\mathcal{H}_{\text{rot}}^{\text{off}} = \widetilde{G}_{12} \cdot (\alpha_1\beta_{21} + \alpha_2\beta_{12})/2$. Phase 1b adds the KH IC factory and a 4-component realizability cone. **Phase 1c closes the D.1 Kelvin–Helmholtz falsifier at the broadband gate:** $\gamma_{\text{measured}}/\gamma_{\text{DR}} = 1.34$ at level 5, mesh-converged to 1.2% between L4 and L5, $n_{\text{neg Jac}} = 0$. The growth is forced kinematic — the closed-loop $\beta_a \leftrightarrow \beta_{\text{off}}$ coupling required for self-amplified Drazin–Reid eigenmode dynamics is documented as a physics extension for follow-up (M4). **Phase 2 closes the D.4 Zel'dovich pancake test:** per-axis γ selectivity verified at $\text{std}(\gamma_1)/\text{std}(\gamma_2) \approx 2.6 \times 10^{14}$ at near-caustic time, γ_1 developing spatial structure (dynamic range 4.18×) while γ_2 stays uniform to round-off, and the Phase 1a strain coupling verified inert ($\max|\beta_{\text{off}}| = 0$) on the axis- aligned IC. **Phase 3 lands the 2D substrate** (`TracerMeshHG2D`, `inject_vg_noise_HG_2d!` with explicit `axes::Tuple` per-axis selectivity, `gamma_per_axis_2d_per_species_field`). **Phase 4 exercises the D.7 dust-trap test** (Taylor–Green vortex + 2-species gas/dust): substrate sound (mass-exact, momentum-exact, γ- separating, NaN-free), but the literal centrifugal-accumulation prediction is FALSIFIED at this milestone — the pure-Lagrangian substrate's variational integrator preserves the multi-species tracer matrix bit-exactly and does not natively capture sub-cell centrifugal drift. Per-species momentum coupling is flagged as an M4+ physics extension. **Phase 5 closes the D.10 ISM multi- tracer fidelity test in the strongest possible form:** `tracers_byte_equal_to_ic == true`, `tracers_max_diff_final == 0.0` across $L \in \{3, 4, 5\}$ under stochastic injection (the 2D analogue of M2-2's structural argument). See `reference/notes_M3_6_phase1c_D1_kh_falsifier.md`, `reference/notes_M3_6_phase2_D4_zeldovich.m` `reference/notes_M3_6_phase4_D7_dust_traps.md`, and `reference/notes_M3_6_phase5_D10_ism_trace`

Milestone 3-7 — 3D extension (DONE; sub-phases a–e). $SO(3)$ Cholesky + Berry coupling on a 3D HG mesh. **M3-7a**: 3D HaloView smoke + 3D field-set allocator (16-named-field `PolynomialFieldSet` on `HierarchicalMesh{3}`; bit-exact round-trip on 64 leaves). **M3-7b**: native HG-side 3D EL residual `cholesky_el_residual_3D!` + Newton driver `det_step_3d_HG!`; the load-bearing dimension-lift gates hold at $\Delta = 0.0$ *absolute* in both §7.1a (3D $\subset$ 1D matches the M1 Phase-1 zero-strain trajectory byte-equal) and §7.1b (3D $\subset$ 2D matches `det_step_2d_HG!` byte-equal). **M3-7c**: SO(3) Berry coupling integration in `cholesky_el_residual_3D_berry` 8 SymPy CHECKs reproduced; iso-pullback ε-extrapolation slope 1 ± 10^{-3}; $\mathcal{H}_{\text{rot}}$ kernel-orthogonality residual $\leq 10^{-10}$. **M3-7d**: per-axis γ + AMR + 3-component realizability projection in 3D; 1D-symmetric selectivity ratio $\text{std}(\gamma_1)/(\text{std}(\gamma_2) + \text{std}(\gamma_3) + \varepsilon) = 6.4 \times 10^{10}$. **M3-7e closes the D.4 3D Zel'dovich pancake** (the cosmological reference test in 3D — the headline 3D scientific success): selectivity ratio 3.0×10^{13} at near-caustic, exceeding the 10^{10} headline gate by three orders. The 3D Cholesky-sector substrate is operational and verified. See `reference/notes_M3_7e_3d_tier_cd_drivers.md`.

Milestone 3-8 — Tier-E + GPU prep (DONE; full GPU port deferred). M3-8a closes Tier E in graceful-degradation mode (E.1 high-Mach Sod, E.2 severe shell-crossing, E.3 low-Knudsen Navier–Stokes limit; +315 asserts; see §10.6). **M3-8b** ships the matrix-free Newton-Krylov drivers (`det_step_{2d, 2d_berry, 3d_berry}_HG_matrix_free!`) using `NonlinearSolve.NewtonRap` `= KrylovJL_GMRES(), concrete_jac = false, jvp_autodiff = AutoForwardDiff()`; bit-equal to round-off vs the dense ForwardDiff-coloured path on every tested IC; CPU-side speedup 1.17–1.87× at levels 3–5 (the algorithm-side prerequisite for the GPU port; sparse-Jacobian construction overhead is removed). **M3-8c (full Metal/CUDA port)** is deferred

pending the upstream HG `Backend`-parameterized `PolynomialFieldSet{<:KA.Backend}`: as of HG `81914c2` the SoA storage is CPU-only, so the per-leaf residual kernel cannot be cleanly ported; the matrix-free driver is shipped as the algorithm-side baseline and the Metal kernel exploration is documented + `@test_skip`-guarded. MPI scaling is also deferred to M4. See `reference/notes_M3_8a_tier_e_gpu_prep.md` and `reference/notes_M3_8b_metal_gpu_port.md`.

Milestone 3-9 — Wrap-up (DONE). Methods-paper honest revision (§10.5 D.1/D.4/D.7/D.10 sub-paragraphs and §10.6 Tier-E paragraph reflect the actual M3-6 / M3-7 / M3-8a results); §11.3 closing milestones table updated to mark M3 substantially closed; M3 final synthesis + M4 plan in `reference/MILESTONE_3_FINAL.md` and `reference/notes_M4_plan.md`.

M3 closure summary. As of M3-9, all six 2D Tier-D headline falsifiers have landed (D.1 KH at the broad-band gate; D.4 Zel'dovich pancake in both 2D and 3D — the headline scientific success of M3; D.7 dust traps with substrate-sound / physics- extension-flagged finding; D.10 ISM multi-tracer fidelity verified bit-exact). Tier E passes in graceful-degradation mode for E.1, E.2, E.3. The 3D Cholesky-sector substrate is operational and verified at the dimension-lift level ($\Delta = 0.0$ absolute) and at the headline-physics level (D.4 selectivity 3×10^{13}). The matrix-free Newton-Krylov solver is bit-equal vs the dense path and 1.17–1.87× faster on CPU. Total test count: $\sim 33{,}940 + 1$ deferred (from 19,687 + 1 at M3-3 close). Open items deferred to M4: D.1 closed-loop β coupling for self-amplified KH; D.7 per-species momentum coupling for sub-cell drift; full Metal/CUDA port (post-HG `Backend`); MPI scaling; E.4 cosmological-with-self-gravity; 3D extensions of D.1, D.7, and D.10.

The design-document Milestones 1–5 map cleanly onto this HG-adapted plan: design-M1 = code-M1; design-M2 = code-M2; design-M3 $\subset$ code-M3-0 through M3-4; design-M4 $\subset$ code-M3-6; design-M5 $\subset$ code-M3-8. The total scope is unchanged from the original design; the engineering unit has shrunk to a single dimension-generic codebase, which is why the 3D extension (originally an outlook item) sits in code-M3 rather than as a separate downstream milestone.

11.4 Outlook

Three natural extensions:

3D extension. Replace $SO(2)$ with $SO(3)$ in the principal-axis decomposition; the structural skeleton is identical. The 6×6 phase-space Cholesky factor has 21 independent components; field count per Lagrangian point reaches ~ 50 including all moments. Octree Eulerian mesh, tetrahedral Lagrangian mesh, r3d for 3D polyhedral overlap. Mostly mechanical from the 2D structure; the genuinely new content is the $SO(3)$ Berry connection.

Cosmological self-gravity. Multigrid Poisson on the Eulerian quadtree, interpolation to Lagrangian vertices. The Zel'dovich pancake test (D.4) requires this; production cosmology needs careful handling of the cosmological-volume periodic boundary conditions and the proper-time vs. scale-factor evolution.

Plasma physics with magnetic field. The framework extends to magnetohydrodynamics by adding the magnetic field as a charge-1 vector under the strain group, with its own connection-form evolution and Israel–Stewart-style relaxation toward force-free or ideal-MHD limits. The closure-quality diagnostics generalize: rank collapse in magnetic field corresponds to current sheet formation, a known difficulty for moment-truncated MHD.

Hybrid kinetic-fluid coupling. The framework's defining feature is honest self-reporting of closure validity. Cells where the diagnostics fire can be flagged for transition to a kinetic solver (PIC, Vlasov, or multi-stream phase-space sheet), with the moment-tower handing off to the kinetic representation seamlessly. The cross-coupling structure for two-fluid systems provides the template for fluid-kinetic coupling.

Acknowledgments

This framework builds on dfmm [1], the r3d polynomial-moment kernel [2], and the Abel-Hahn-Kaehler phase-space sheet formulation [3]. The variational structure was developed in iterative working notes (v1, v2, v3) with checkpoint feedback at each stage; the v3 empirical findings on gamma-distributed burst durations and the principal-axis matrix lift are due to the working-note iteration on dfmm's existing wave-pool calibration data.

References

[1] T. Abel, "A Lagrangian-coordinate-aware moment scheme with built-in closure diagnostics for moderate-Knudsen flows," *In preparation*; code repository `https://github.com/yipihey/dfmm`.

[2] D. M. Powell and T. Abel, "An exact general remeshing scheme applied to physically conservative voxelization," *J. Comput. Phys.* **297** (2015) 340–356; code repository `https://github.com/devonmpowell/r3d`.

[3] T. Abel, O. Hahn, R. Kaehler, "Tracing the dark matter sheet in phase space," *Mon. Not. R. Astron. Soc.* **427** (2012) 61–76.

[4] Working note v3 (this design document), April 2026.

[5] T. Abel, "`HierarchicalGrids.jl`: a dimension-generic hierarchical mesh, polynomial-field, and remap substrate in Julia," v0.1, April 2026; code repository `https://github.com/yipihey/HierarchicalGrids.jl`.

[6] T. Abel, "`r3djl`: Julia port of the r3d polynomial-moment kernel," April 2026; code repository `https://github.com/yipihey/r3djl`.

[7] T. Abel, "HierarchicalGrids.jl design guidance for dfmm-2d," working note, April 2026; reproduced as `reference/notes_HG_design_guidance.md` in the dfmm repository.

[8] A. Hill and G. Dalle, "`SparseConnectivityTracer.jl`: sparse Jacobian and Hessian connectivity detection in Julia," *J. Open Source Softw.*, code repository `https://github.com/adrhill/SparseConnectivityTracer.jl`.